

Data-driven estimation of forest carbon stocks in Spain: a machine learning approach using National Forest Inventory data

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Abstract

Forests are natural sinks of CO_2 and their management is acquiring a central role in emission compensation regulatory frameworks and in voluntary carbon credit markets. However, the reliable estimation of the carbon that a forest plot will accumulate over a given time horizon remains a methodological challenge, especially when temporal projection from initial conditions is required. This work presents *GreenWest*, a machine learning system to predict future forest carbon in Spain from stand structure, soil, climate and remote sensing data. Data come from the National Forest Inventory (IFN2 and IFN3), ERA5-Land climatic reanalysis, Landsat spectral indices and edaphic and topographic variables, integrated in a multi-source relational database. Through manual feature selection, an initial set of 445 predictors was reduced to 44 ecologically relevant variables, modelling two target variables: carbon density (c_4 , $tC\ ha^{-1}$) and total plot carbon ($carbono_bruto_4$, tC). *Gradient boosting* models (LightGBM, XGBoost, CatBoost) and neural networks were trained and compared as base models, as well as stacking ensembles with different meta-model configurations. The best results are obtained with LightGBM and CatBoost (R^2 between 0.72 for IFN2/ c_4 and 0.90 for IFN3/ $carbono_bruto_4$), with systematic but modest improvements when incorporating stacking. Models trained on the combined IFN2+IFN3 set outperform local models on IFN2, while for IFN3 the opposite occurs. SHAP analysis reveals that tree density, species and canopy cover fraction are the dominant predictors, with the summer GNDVI index as the main spectral predictor; SHAP values are consistent with ecological knowledge, validating the model's capacity to learn relationships with genuine meaning.

1. Introduction

Climate change is one of the greatest global challenges and its most direct manifestation is the increase in atmospheric concentrations of carbon dioxide (CO_2), with impacts on the cryosphere, climate extremes and ecosystems [1]. Forests act as natural sinks by fixing CO_2 in biomass via photosynthesis, making their management key for mitigation.

Over recent decades, international instruments such as the *United Nations Framework Convention on Climate Change (UNFCCC)* and the *Kyoto Protocol* [2, 3] have established regulatory frameworks for reducing greenhouse gas emissions through market-based mechanisms. In this context, *carbon credits* arise: units representing the amount of carbon dioxide (CO_2), usually one tonne, that has been captured or whose emission has been avoided through certified mitigation projects.

Among eligible activities, afforestation and reforestation stand out for their capacity to act as natural carbon sinks, fixing CO_2 in biomass and soil. However, for these actions to generate valid carbon credits they must meet a series of technical and legal criteria established in international climate change regulations and in their national application. In particular, these requirements derive from the forest sink

accounting rules adopted under the United Nations Framework Convention on Climate Change (UNFCCC) and the Kyoto Protocol, as specified in the Marrakech Accords, as well as from the national forest definition communicated by Spain for these purposes [4, 5]. These criteria include:

- **Direct human intervention:** The trees must result from direct human intervention activities such as planting, seeding or the deliberate promotion of natural regeneration. This requirement derives from the definition of *afforestation* and *reforestation* established in the Kyoto Protocol, which expressly excludes natural regeneration not induced by human action [4].
- **Minimum permanence period:** The project must ensure the permanence of the carbon sink for a prolonged period (usually on the order of 20–30 years), in order to guarantee that the captured carbon is not prematurely released into the atmosphere. This criterion responds to the permanence principle required in the LULUCF forest sink accounting regime and in national and European implementation frameworks, which excludes short-rotation crops whose carbon is released after harvest [4, 6].
- **Minimum area of 1 hectare:** The project area must have a minimum extent of 1 hectare. This threshold comes from Spain's national forest definition adopted within the ranges permitted by the Marrakech Accords (0.05–1 ha), officially communicated to the UNFCCC [5].

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- **Minimum canopy cover fraction of 20%:** For land to be considered forest, the crown cover of tree species must reach at least 20% of the surface area. This value corresponds to Spain's national choice for the forest definition for climate accounting purposes [5].
- **Minimum height of mature trees of 3 metres:** Tree species must be capable of reaching a minimum height of 3 metres at maturity. It is not necessary for this height to be reached at the start of the project, but it must be attainable under normal growth conditions. This criterion is also part of Spain's national forest definition communicated in accordance with decisions adopted under the UNFCCC [5].

This work aims to obtain an artificial intelligence model to estimate the amount of carbon that a forest plantation in Spain will capture from vegetation, climate and terrain variables over a period of 20 to 30 years. This innovative approach has the potential to transform the management of afforestation and reforestation projects, optimising planting practices and maximising the amount of carbon that can be captured in these ecosystems.

The operational question is: *given the initial characteristics of a plantation, how much CO_2 will it contain after t years?, where t is a natural number.* To answer it, data from the **National Forest Inventory** (IFN2–IFN4, MITECO) [7], **ERA5-Land** reanalysis [8] and **Landsat spectral indices** (Collection 2, L2) [9] are integrated into a hierarchical relational database described in a companion work [10].

This model will not only improve understanding of the behaviour of carbon sinks, but will also provide useful tools for strategic decision-making in both business and environmental spheres. In this way, the *GreenWest* project contributes to the transition towards a low-carbon economy, aligning with the global sustainability objectives established within the framework of the UNFCCC and the *Kyoto Protocol*, and promoting the creation of a more efficient and accessible carbon credit market for the economic actors involved in the management of natural resources.

2. Objectives and Justification

The main objective of this study is to develop an artificial intelligence model capable of accurately predicting the capacity of Spanish forest plantations to absorb carbon dioxide (CO_2). This model is based on variables describing tree species, terrain characteristics and climatic and meteorological conditions. From this general objective, several specific goals are derived, which together justify the relevance and applicability of the project.

2.1. Specific objectives

The main objectives are listed below:

- **Develop a robust predictive model:** Build a machine learning model that estimates the amount of CO_2 that will be captured over time by a forest plantation, based on data such as species, soil type, diameter class, climate and other relevant variables.
- **Ensure compliance with international regulations:** Guarantee that the model's predictions and outputs are compatible with the regulatory frameworks defined by the *United Nations Framework Convention on Climate Change* (UNFCCC) and the *Kyoto Protocol*, thus meeting the criteria required for the validation of carbon credits.
- **Analyse the determining factors of forest development:** Study the influence of climatic variables (such as temperature and precipitation) and edaphic variables (such as soil type or slope) on forest growth and its carbon capture capacity.
- **Support environmental and business decision-making:** Provide a practical and validated tool that enables technicians, managers and companies to select the most suitable species and plan afforestation actions with maximum efficiency in terms of carbon sequestration.

2.2. Justification

The need for predictive tools to estimate CO_2 capture has intensified given the growth of the voluntary carbon credit market, and the obligations acquired: each country must report its greenhouse gas emissions and absorptions, and can use (re)forestation activities as offsetting mechanisms.

For these projects to be eligible, they must meet specific criteria, which make it essential to have models that not only estimate current carbon but are capable of forecasting its future evolution based on initial conditions and predictor variables.

This work seeks to fill that gap through the use of artificial intelligence applied to real, multi-source data. Integrating its management within the carbon credit system can represent an important opportunity for the local economy, which could support those regions where depopulation and the demographic challenge are a serious problem, and for the mitigation of climate change.

3. Literature Review

The precise quantification of forest resources has historically been one of the cornerstones of territorial management and natural resource economics. The evolution of techniques for measuring tree growth and, more recently, for estimating biomass and carbon content, reflects a profound transformation in human society's priorities with respect to forest ecosystems. What began in medieval Europe as a logistical need to ensure the supply of firewood and structural timber in the face of scarcity has metamorphosed in the 21st century into a high-technology scientific discipline driven by global climate urgency [11].

Traditional dendrometry, the pillar of modern forest inventories, is based on the use of allometric relationships to estimate biomass (w) and other ecological parameters from easily measurable field variables, mainly diameter at breast height (D) and total height (H). These estimates are usually

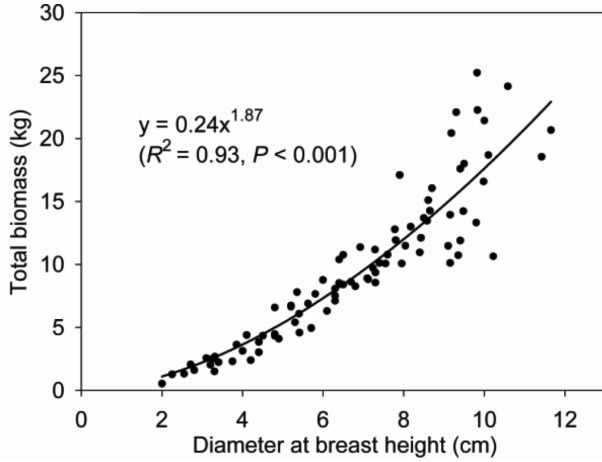


Figure 3.1: Relationship between total biomass in kilograms and diameter at breast height in centimetres for moso bamboo (*Phyllostachys edulis*). Seen in [12], with [14] as the original source.

articulated through equations of the form:

$$w = aD^bH^c \quad (3.1)$$

or their logarithmic transformations:

$$\lg w = a + b \lg D + c \lg H \quad (3.2)$$

where parameters a , b and c are empirical regression coefficients [12]. In this context, the precision of primary measurements is critical, since, given the potential nature of these functions, errors in measuring D and H propagate and amplify in the final calculation of volume and carbon content. Despite the apparent simplicity of the fitting, results can be very good, as shown in Figure 3.1. However, there is a tension in the literature between the use of “pantropical” or generalised equations and species- or site-specific equations, since biases can be introduced if the architecture of local trees differs from the global one [13].

The measurement of tree height has historically posed a greater challenge than that of diameter. Until the 1990s, mechanical hypsometers based on trigonometry predominated, which required manually measuring the distance to the tree and a clear line of sight [15]. These methods suffered from limitations of accuracy and ergonomics. The introduction of electronics marked a turning point by using ultrasound to measure distances automatically, enabling work in dense understoreys and improving precision to below 1% [15]. At the same time, modern electronic callipers (an instrument measuring the linear distance between two parallel tangents to the tree stem) have digitised data collection, integrating measurement and metadata recording to minimise transcription errors [15].

Subsequently, the introduction of laser scanning technology represented a revolution in forest measurement, overcoming the logistical and accuracy limitations of traditional methods. This technology is deployed mainly in two modalities: terrestrial laser scanning (TLS), which captures the

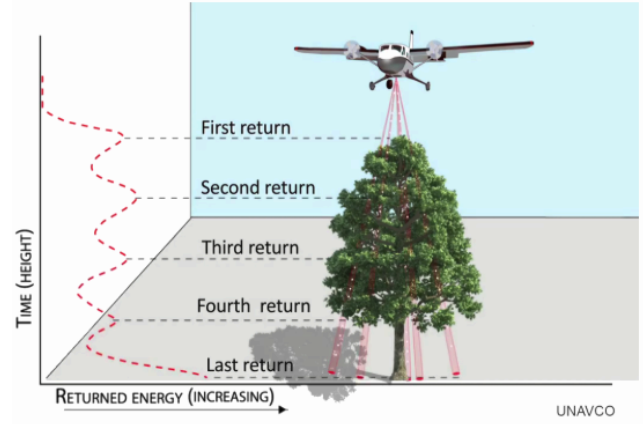


Figure 3.2: Diagram of a tree being scanned by an airborne LiDAR system and the various reflected pulses. Image obtained from [17].

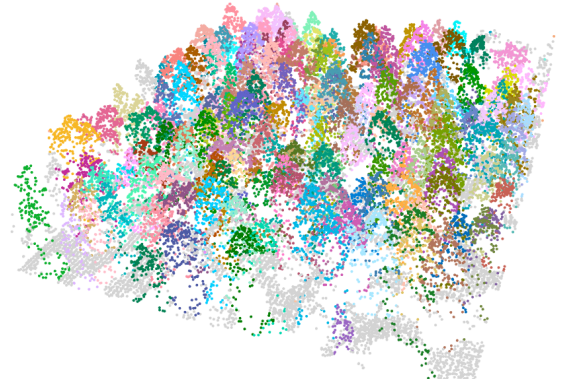


Figure 3.3: Visualisation of a forest scanned with LiDAR. The possibilities of identifying individual trees in a dense forest environment are shown [18].

three-dimensional structure of the forest from the ground with millimetre detail [16], and airborne LiDAR. The latter is a method that uses light pulses to measure distances. Large-scale surveys typically consist of a LiDAR system mounted on an aircraft, helicopter, drone or satellite, which measures the distance to objects below it. One advantage is that, just as light can reach the ground through gaps in tree crowns, LiDAR laser pulses do so too, as we can observe in Figure 3.2. This means that in a single overflight various layers at different heights can be captured, enabling 3D reconstructions of the scanned area. The level of detail achievable is so high that individual trees can be identified, as seen in Figure 3.3.

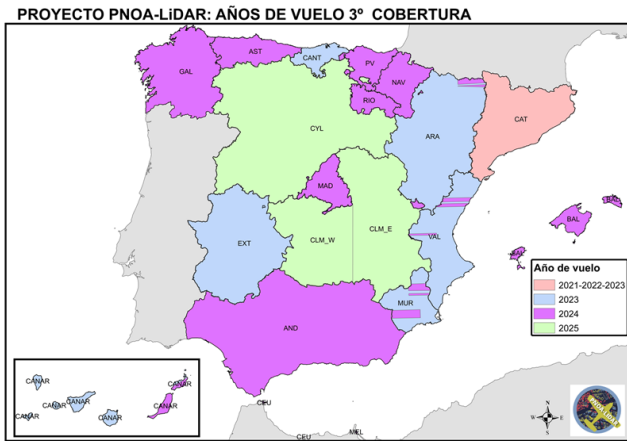


Figure 3.4: Data acquisition plan for the Third cycle of the PNOA-LiDAR project of Spain's National Geographic Institute [19].

The main limitations of airborne LiDAR lie in its high cost and logistical complexity. These factors hinder large-scale coverage with optimal periodicity, also generating significant latency in the availability of regional data. Furthermore, processing this information is more demanding than previous methods, compounded by the technical challenge of managing and storing large volumes of data. Figure 3.4 shows the data acquisition plan for the Third cycle of the PNOA-LiDAR project of Spain's National Geographic Institute, where the cadence of mappings can be appreciated.

The integration of these structural data, together with satellite remote sensing and the large-volume data processing capability offered by machine learning algorithms, has inaugurated a new paradigm: the ability to monitor forest resources at global scale with unprecedented precision. It is in this context of “precision forestry” and large-scale Earth observation that current research is situated, with very promising results having been achieved that enable the complexity of forest ecosystems to be addressed with a granularity previously unattainable.

4. State of the Art

Carbon sequestration in forest ecosystems has gained growing importance in the scientific literature, driven both by international commitments on climate change and by the rise of carbon credit markets. This has motivated the development of models aimed at quantifying forest biomass and estimating carbon content, taking advantage of recent advances in remote sensors and artificial intelligence techniques.

One of the most established strategies for quantifying forest carbon is the estimation of stored carbon at a given moment from remote sensing data. Goetz et al. (2009) [20] review the use of satellite observations, including optical sensors such as MODIS and Landsat, in empirical models of aboveground biomass, highlighting their applicability at regional scale, especially in boreal ecosystems. These

types of estimates are usually based on linear regressions or generalised least squares models, with coefficients of determination typically between 0.6 and 0.8, depending on spatial resolution and ecosystem heterogeneity.

The application of deep learning has substantially improved the accuracy and spatial resolution of these estimates. For example, Zhang et al. (2022) [21] integrate Sentinel-2 imagery with convolutional neural networks, achieving an R^2 of 0.84 for estimating carbon in subtropical forests. Similarly, Jiang et al. (2022) [22] develop the *ForestCarbonAI* model, trained with multispectral and LiDAR data, generating high-resolution (10 m) forest carbon maps and reporting mean absolute errors (MAE) below 3.5 tC/ha in temperate zones. Other recent works, such as Reiersen et al. (2022) [23] or Dong et al. (2023) [24], also demonstrate the effectiveness of *deep learning* for static estimates, although they focus on tropical contexts and do not consider the temporal component.

In contrast to these descriptive approaches, some initiatives have attempted to project the evolution of carbon into the future. At the national level, the Ministry for Ecological Transition (MITECO) has implemented tools such as the *ex ante* carbon absorption calculator [25], which provides simplified estimates of the carbon that can be fixed in a forest plantation depending on species and agroclimatic zone. However, this instrument is based on tabulated coefficients and does not incorporate real edaphoclimatic variables or data-based modelling techniques, which limits its accuracy and ability to adapt to specific contexts.

At the European level, the work of Fasihi et al. (2024) [26] stands out, applying an approach similar to that proposed in this study in the Friuli-Venezia Giulia region (Italy). Their objective is to estimate both the current carbon *stock* and its annual absorption rate (sequestration), based on dendrochronology data from the Italian National Forest Inventory. These measurements, carried out between 2017 and 2019, quantify the radial growth of the last five years to estimate biomass through allometric equations and its conversion to carbon according to IPCC guidelines. They then train predictive models using meteorological, geomorphological variables, vegetation indices and a LiDAR canopy height model (CHM), attempting to predict the values obtained from field data.

The authors evaluate various ensemble models, reporting that combinations of algorithms such as Random Forest, AdaBoost and CatBoost offer the best performance. In predicting carbon *stock*, they achieve an R^2 of 0.73 ± 0.07 and an RMSE of 31.55 ± 9.35 tC/ha. For the sequestration rate, results are more modest, with an R^2 of 0.42 ± 0.08 and an RMSE of 0.90 ± 0.08 tC/ha/year. A key finding of the study is that the inclusion of LiDAR data dramatically improves the accuracy of estimates.

In this scenario, the present work proposes an innovative methodology focused on dynamic long-term carbon prediction. Unlike previous models, which estimate already-stored carbon, this study focuses on anticipating how much carbon a forest plantation will capture over a specific time

horizon. To this end, various supervised learning models are studied, trained with historical data from the National Forest Inventory (IFN2, IFN3 and IFN4) [7] including edaphic characteristics, climatic variables from Copernicus [27, 28] and spectral metrics derived from Landsat images [29]. Details on the model architecture, variables used, algorithms implemented and evaluation metrics are developed in the following sections.

5. Methodology

This section describes the procedure followed for training and validating the predictive models. The methodology is grounded in the factors that determine forest growth: meteorological conditions (temperature, precipitation) affect photosynthesis and water availability; aspect, slope and altitude modulate incident radiation and local microclimate; and tree density determines intraspecific competition [30]. The approach integrates structural, climatic and spectral information from the IFN and complementary environmental sources to predict carbon accumulated in living biomass.

5.1. Data origin and structure

The database integrates forest, climatic and spectral information at the plot, species and diameter class level, structured around the IFN2, IFN3 and IFN4 inventories. In addition to the IFN dendrometric variables, each plot incorporates seasonal climatic statistics (temperature, precipitation, Martonne index) and spectral indices derived from satellite imagery (NDVI, EVI, NDII, GNDVI). Figure 5.1 shows the relational schema of the main tables; a complete variable dictionary is given in [10].

5.2. Target variables

The goal of the model is to estimate the **total carbon** that a forest plot will capture over a time horizon of 20–30 years, based on conditions observed in previous inventories. Two complementary response variables are considered, both derived from the National Forest Inventory (IFN) data, allowing carbon content to be analysed from different perspectives: one normalised by area and another in absolute terms.

1. **c (tC/ha)**: represents the **total carbon contained in aboveground and belowground living biomass** per unit area, expressed in *tonnes of carbon per hectare*. Its calculation is based on the sum of the aboveground (ca) and root (cr) carbon estimates reported by the IFN. In cases with missing values, the information was completed using a *Random Forest Regressor* model fitted on observed dendrometric variables (Species, CD, VSC, NPies, ABas, IAVC, VCC and VLE), achieving satisfactory performance ($R^2_{test} > 0.90$). This variable is consistent with international forest inventory reporting formats and allows carbon content to be compared across plots or species.
2. **carbono_bruto (tC)**: corresponds to the **total carbon captured per plot and species**, expressed in *total tonnes of carbon*. Its estimation is performed in a

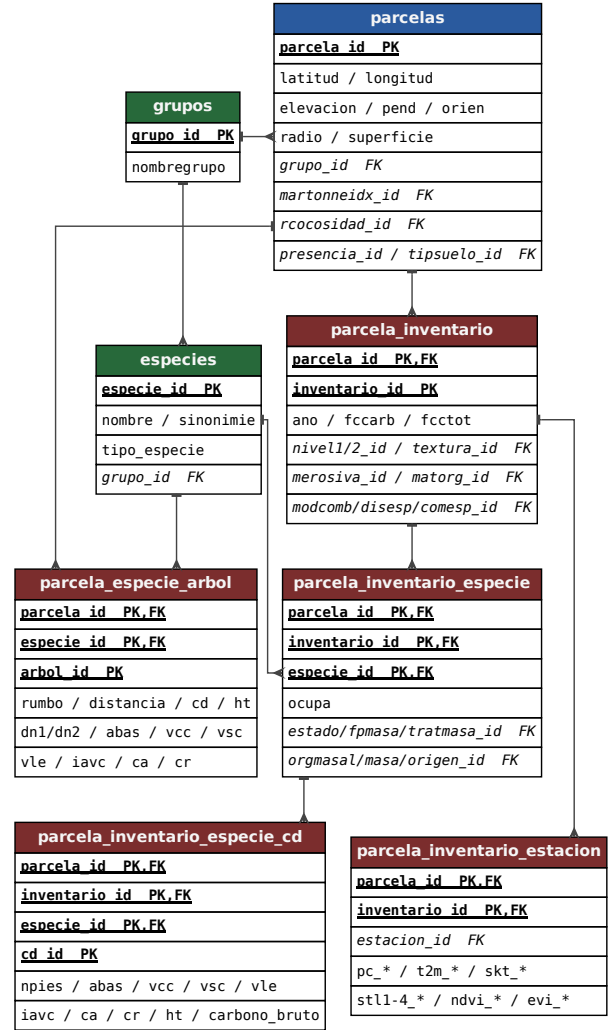


Figure 5.1: Relational schema of the main database tables. Figure extracted from [10], where further details on the variables can be found.

traceable and physically interpretable manner from variables measured directly in the field: number of trees (npies), mean height (ht), species type (clase_especie) and diameter class (cd_id). The calculation follows an allometric model adapted from [31] and the IPCC guidelines [30], incorporating both aboveground and root biomass via the Root-to-Shoot ratio (R). The result is expressed in total tonnes of carbon per plot, without normalisation by area, which facilitates the traceability of the process and the comparison between inventories without depending on IFN-specific expansion factors. Consistent with the criteria for afforestation and reforestation projects, observations corresponding to seedlings or saplings are assigned a zero carbon value, since early development stages are not officially counted as captured carbon.

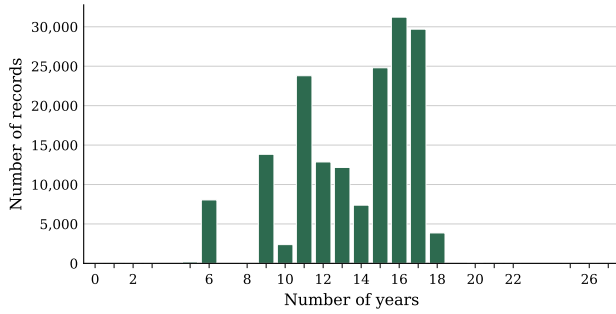


Figure 5.2: Distribution of the year difference between the third and fourth inventories.

These two variables summarise forest carbon content from complementary perspectives: c (tC/ha) enables spatial and temporal comparison between forest stands, while carbono_bruto (tC) provides an absolute measure directly derived from field observations. Both constitute the main targets of the predictive modelling, aimed at estimating the carbon accumulated in **IFN4** from the conditions recorded in the previous inventories (**IFN2** and **IFN3**).

For further details on the origin of these variables, see [10].

5.3. Eligibility assumptions and external verification

As already introduced, for a forest project to be eligible for *carbon credit* programmes in Spain, certain technical requirements must be met [30, 32]. Each criterion and the way it is addressed in this study are summarised below:

- **Direct human intervention.** The carbon increment must result from planned actions (reforestation, restoration or sustainable management). In our case, the model is trained on observational data (**IFN2**–**IFN3**–**IFN4**); therefore, *intervention verification* is not inferred from the model, but is considered as an *external condition* of eligibility for the project to be evaluated.
- **Minimum permanence.** The interval between **IFN3** and **IFN4** is mostly between 6 and 17 years (Figure 5.2), a range too narrow for the 20–30 year reference horizon. To extend temporal coverage, the **IFN2**–**IFN3** pair was also integrated, yielding intervals of 6 to 29 years (Figure 5.3), more representative of the target horizon.
- **Minimum area of 1 ha.** This is a criterion external to the model, verifiable *ex ante* from the declared project geometry. In homogeneous stands, total carbon is approximately proportional to area, so this requirement does not affect model fitting.
- **Minimum canopy cover fraction of 20%.** The database provides fccarb (tree canopy cover) and fctot (total cover). This threshold is applied as an *eligibility filter* prior to modelling ($\text{fccarb} \geq 20$).
- **Minimum height of 3 m at maturity.** This refers to the adult height of the species, not to the seedlings at the time of inventory. It is evaluated externally

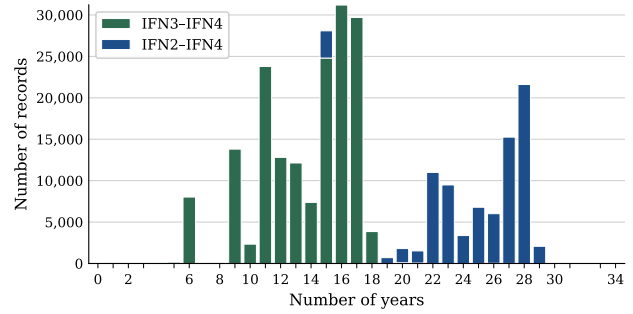


Figure 5.3: Distribution of the year difference between the **IFN2**–**IFN3** and **IFN3**–**IFN4** inventories.

through the selection of appropriate species and does not intervene in model training.

5.4. Data preparation and processing

As already introduced, training is carried out along two lines according to the target variable: carbon in tonnes per hectare (c from **IFN4**) or carbon in tonnes (carbono_bruto from **IFN4**); and according to the information used as explanatory variables: **IFN3** or **IFN3** and **IFN2**. Data preparation and filtering are presented in general terms.

5.4.1. Record filtering

All plots in which the total carbon value (target variable) at the second inventory is lower than at the first are discarded. These cases are usually due to deforestation episodes, fires or other disturbances, and do not represent net forest growth.

The dataset is restricted to plots with a fccarb (tree canopy cover fraction) equal to or greater than 20 % in the explanatory inventory. This threshold defines the minimum proportion of area occupied by tree crowns relative to the total plot area, and constitutes one of the essential conditions for considering a surface as forest land. Excluding plots with fccarb below 20 % ensures that carbon estimates are made on consolidated forest stands, avoiding biases associated with agricultural areas or shrublands. The **IFN2** data are not subject to this filter because the fccarb variable is not available for that inventory.

5.4.2. Variable calculation and aggregation

Each input record is generated at the plot–species combination level, incorporating the corresponding variables from the first measurement and the target variable (carbon) from the second measurement (**IFN4**). The parcela and $\text{parcela_inventario}$ variables are split for each species. The entries in the $\text{parcela_inventario_especie_cd}$ table are grouped by plot and species and compressed into a single entry by creating a set of variables for each diameter class.

5.4.3. Reclassification of the pendiente and orientacion variables

Since the ecological effect of slope and aspect on carbon accumulation is non-linear, both variables were reclassified

as categorical factors. The `pendiente_cat` variable was defined using seven intervals ($< 5^\circ$, $5-10^\circ$, $10-15^\circ$, $15-20^\circ$, $20-30^\circ$, $30-50^\circ$, $> 50^\circ$); the `orientacion_cat` variable into eight equiangular cardinal sectors (N, NE, E, SE, S, SO, O, NO).

5.4.4. Grouping of the periodo variable

As can be seen in Figure 5.3, the distribution of the `periodo` variable, defined as the number of years elapsed between the measurement of the explanatory variables and the observation of the target variable, shows some heterogeneity in its frequency. Although the total range of values extends approximately between 0 and 34 years, some temporal intervals are represented by a very small number of observations.

This scarcity of data for certain values of `periodo` can introduce instability in model training, by forcing the algorithm to learn patterns from poorly representative samples. To mitigate this effect and improve the robustness of predictions, certain values of `periodo` were grouped into broader temporal categories, giving rise to a new variable called `periodo_agrupado`.

The grouping procedure was designed conservatively, keeping unchanged those values with sufficient sample support and grouping only the most scarce intervals. Specifically, values below 6 years were grouped into category 5; periods between 7 and 10 years were grouped into 10; those between 18 and 21 years were grouped into 20; and values equal to or greater than 29 years were grouped into 30. The remaining intermediate values were left unchanged.

It should be noted that the `periodo_agrupado` variable does not retain the full granularity of the original variable, but does retain its essential temporal meaning. The experiments carried out show that this representation is more statistically stable and leads to models with more robust predictive behaviour, by reducing sensitivity to temporal intervals with low observation frequency.

5.5. Partitioning and validation

To obtain an unbiased estimate of performance and avoid *data leakage* arising from the hierarchical structure of the data, the training process is organised at two levels: (i) an external *hold-out* partition for final evaluation and (ii) an internal cross-validation for hyperparameter selection.

Train/test split. The filtered data are split into 80% for training and 20% for test. This split is made so that all observations associated with the same plot are assigned entirely to one of the subsets.

Cross-validation for hyperparameter selection. Hyperparameter selection is carried out exclusively on the training set using `GridSearchCV` with R^2 as the optimisation metric. `GroupKFold` with $k = 5$ folds is used, ensuring there is no overlap of plots between folds (i.e., the grouping is respected both in the hold-out and in the internal validation).

Evaluation metrics. The main metrics used are RMSE, RMSE%, R^2 , MAE and SMAPE. In the following expressions, y_i is the observed value, $\hat{y}_i$ the predicted value, $\bar{y}$ the observed mean and n the number of observations:

- **RMSE:** root mean squared error, in the same units as the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5.1)$$

- **RMSE%:** RMSE expressed as a percentage of the observed mean.

$$\text{RMSE}_{\%} = \frac{\text{RMSE}}{\bar{y}} \times 100 \quad (5.2)$$

- **MAE** (mean absolute error): average of absolute errors; more robust than RMSE against outliers by penalising all residuals equally.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5.3)$$

- **R^2 (coefficient of determination):** proportion of variance explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (5.4)$$

- **SMAPE:** symmetric percentage error, bounded and robust against the asymmetry of MAPE.

$$\text{SMAPE} = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \quad (5.5)$$

As supporting metrics for residual analysis and bias detection, MAE, MAPE, wMAPE, MedAE and Bias are also calculated.

5.5.1. Encoding and normalisation

In order to ensure methodological consistency and avoid *data leakage* during cross-validation, all preprocessing steps are explicitly integrated inside a Pipeline together with the regression model. In this way, the parameters associated with preprocessing are estimated *exclusively* from the training data of each fold, and subsequently applied to the validation or test data.

Variables are treated according to their nature:

- **Continuous numerical variables:** imputed using the median to reduce the influence of extreme values and subsequently standardised via *z-score* normalisation (zero mean and unit standard deviation). This step is especially relevant for models sensitive to variable scale, such as linear regressions, SVR or neural networks.
- **Density structural variables** (`npies_*`): as they represent counts per diameter class, they are imputed with zero when absent and standardised in the same way as numerical variables, preserving their relative contribution in the model.

- **Categorical variables:** imputed using the mode, explicitly converted to string type and encoded via *one-hot encoding*. The `handle_unknown='ignore'` option is used to ensure the model's robustness against categories not observed during training.

This strategy ensures that the encoding, imputation and scaling of variables are performed consistently across all evaluated models and that the estimated performance faithfully reflects the system's generalisation capacity, without introducing biases derived from improper access to information from the evaluation set.

5.6. Feature selection

Predictor selection combined automatic methods and expert criteria. Among the automatic methods, *Featurewiz* was used, which eliminates predictors with high correlation ($|r| > 0.70$) and refines the selection via *Gradient Boosting*, and *mRMR* (minimum Redundancy–maximum Relevance), which maximises the relevance of each predictor with respect to the target while minimising redundancy among them via mutual information.

These results were integrated with a manual selection based on statistical (correlations, ANOVA, redundancy analysis) and ecological criteria, guaranteeing the representation of the essential dimensions of the system (stand structure, topography, soil, climate and spectral indices). Additionally, a supervised sequential procedure (SSSRP) of *forward selection* was applied on a CatBoost model, incorporating at each iteration the variable with the greatest gain in R^2 until $\Delta R^2 < 10^{-5}$ was reached. The final result was a set of 44 variables selected from 445 candidates.

5.7. Evaluated models

Nine representative algorithms from different paradigms were evaluated: *boosting* methods (XGBoost, LightGBM, CatBoost, GBDT), *bagging* methods (Random Forest, BaggedDT) and reference models (SVR, MLP, BayesianNN). Their main characteristics are summarised in Table 5.1. The best base models were subsequently combined in *stacking* configurations.

5.7.1. Stacking configuration

After evaluating all the above models, different base learner configurations were built that are combined via a meta-model. These combinations were designed according to two main criteria:

1. **Structural diversity:** mixing boosting and bagging methods, as well as boosting variants with different growth strategies and regularisation.
2. **Individual performance:** preferentially including models with higher R^2 and lower error (RMSE, MAE) in individual tests.

The meta-models used to integrate the predictions were:

- **Linear models:** Linear Regression, Ridge.
- **Tree-based models:** Random Forest, Gradient Boosting Regressor.

- **Kernel models:** Linear SVR.
- **Neural network:** MLP with one hidden layer.

This selection allows comparison from simple linear combiners to non-linear integrators capable of capturing complex interactions between predictions.

5.7.2. Model comparison and justification

The exhaustive evaluation of multiple algorithms allows identifying not only the best-performing individual model, but also synergistic combinations for *stacking*. Table 5.1 summarises the models finally trained and evaluated.

6. Pipeline Implementation

The development and evaluation of the predictive models were carried out entirely in **Python**, using *scikit-learn*, *cuML*, *PyTorch* and the *gradient boosting* libraries *XGBoost*, *LightGBM* and *CatBoost*. Models trained exclusively with IFN3 data were run on a local machine (Intel Core i7, 32 GB RAM), while those using IFN2 and IFN3 data jointly were trained on the high-performance computing (HPC) system of the University of Salamanca, leveraging Nvidia H100 graphics cards to accelerate the training of GPU-compatible models.

6.1. Feature selection

6.1.1. Manual selection

The manual feature selection started from a thematic organisation of the predictor set, grouping variables according to the type of ecological, structural or climatic information they represent. This classification allowed the dimensionality reduction process to be structured around the following conceptual blocks:

- **Fixed variable block:** describes the basic structure of the forest stand and the essential identification and general characterisation attributes of each plot.
- **Species variable block:** collects information on the composition, state and specific characteristics of the forest formations.
- **Substrate block:** integrates edaphic and management variables susceptible to change over time.
- **Terrain block:** groups physical properties of the medium that remain stable at the temporal scale of inventories (slope, aspect, soil type, etc.).
- **Summary climatic block:** represented by the Martonne aridity index, which synthesises the interaction between temperature and precipitation.
- **Detailed climatic block:** includes explicit seasonal metrics for temperature and precipitation.
- **Vegetation index block:** collects spectral information related to water status, vigour and photosynthetic activity of the vegetation.

In total, the database initially contained 445 candidate variables distributed among these thematic blocks. After

Model	Type	Characteristics	Notes
Random Forest	Bagging	Bootstrap with random feature selection	Robust and stable
BaggedDT	Bagging	Unpruned trees on bootstrap samples	Improvement by aggregation
XGBoost	Boosting	L1/L2 regularisation, second order	Very accurate; sensitive to tuning
LightGBM	Boosting	Leaf-wise growth, very efficient	Fast; risk of overfitting
CatBoost	Boosting	Ordered encoding; robust to noise	Excellent without extensive tuning
GBDT	Boosting	Sequential trees fitted to residuals	Good performance
MLP	Neural network	Captures non-linear relationships	Requires normalisation
SVR	Margin-based	Linear kernel, large margin	Robust against overfitting
BayesianNN	Probabilistic	Quantifies uncertainty	Reduces overfitting

Table 5.1

Supervised learning models evaluated.

applying the manual selection procedure, supported by statistical, ecological and model performance comparison criteria, the set was reduced to 44 representative variables. The variables finally selected within each block were:

- **Fixed variable block:** especie_id, tipo_especie, grupo_id, periodo, radio, ocupa, npies_1, npies_2, npies_5, npies_10, npies_15, npies_20, npies_25, npies_30, npies_35, npies_40, npies_45, npies_50, npies_55, npies_60, npies_65, npies_70.
- **Species variable block:** estado_id, fccarb, disesp_id.
- **Substrate block (dynamic):** modcomb_id, nivel2_id, tratmasa_id.
- **Terrain block:** rocosidad_id, orientacion_cat, elevacion, pendiente_cat.
- **Summary climatic block (Martonne):** martonneidx_id.
- **Detailed climatic block (temperature and precipitation):** skt_mean_primavera, skt_mean_verano, skt_std_primavera, skt_std_verano, pr_sum_invierno, pr_sum_otoño, pr_sum_primavera, pr_sum_verano.
- **Vegetation index block:** gndvi_mean_verano, ndii_mean_primavera, gndvi_std_primavera, evi_mean_primavera.

The comparison of models trained with incremental combinations of blocks showed that all of them contribute relevant information, following the approximate order of contribution: *fixed variables* > *species variables* > *substrate* > *terrain* > *vegetation indices* > *Martonne* > *temperature and precipitation*. That is, most of the predictive capacity is explained by the structure and composition of the forest stand, while edaphic, topographic and climatic conditions act as additional modulators of carbon accumulation.

6.1.2. Automatic selection

Featurewiz selected **67 variables**, with preference for vegetation indices derived from Sentinel-2 (NDII, EVI, GNDVI and NDVI, with mean, maximum, minimum, median and standard deviation statistics, especially in spring and summer) and seasonal thermal and precipitation variables (t2m_*, skt_*, stl_*, pr_sum_*, pr_max_*, pr_min_*). The Martonne aridity index and a representative set of structural variables (trees per diameter class), species and terrain variables completed the selection.

Table 6.1

Stacking base model configurations.

Config.	Base models
1	LightGBM, Random Forest
2	LightGBM, XGBoost, GBDT
3	CatBoost, Random Forest, GBDT
4	CatBoost, LightGBM, Random Forest, GBDT
5	CatBoost, LightGBM, XGBoost, Random Forest, GBDT, BaggedDT

mRMR selected **50 variables** prioritising high mutual information with carbon and low internal redundancy. The set integrates structural predictors (species, radius, diameter classes, aspect and slope), edaphic variables (rockiness, soil types), seasonal climatic metrics and seasonal means of NDII, GNDVI and EVI, confirming that photosynthetic activity and water status act as direct predictors of stored carbon.

6.1.3. Selection result

Of the three selected variable sets, the manual selection was retained as it demonstrated better performance with greater simplicity.

6.2. Stacking configuration

Aggregation via *stacking* was implemented with out-of-fold (OOF) predictions generated on the training set with the same GroupKFold scheme; each base model was retrained on the entire training set to produce the final predictions on test. The meta-variables were standardised using StandardScaler before fitting the meta-model.

With the aim of studying the trade-off between ensemble diversity, computational cost and performance, five base model configurations were defined (Table 6.1). BayesianNN, SVR and MLP models were discarded as candidates due to their lower individual performance.

The configurations range from the lightest (Config. 1, two models with different paradigms) to the most complete (Config. 5, all models with competitive performance), allowing the effect of structural diversity and computational cost

Table 6.2Meta-models used in *stacking* together with their parameters.

Meta-model	Parameters
Gradient Boosting	Default configuration
Linear Regression	No regularisation
Ridge	L2 regularisation with cross-validation ($\alpha \in \{0.01, 0.1, 1, 10, 100\}$)
Random Forest	50 trees
SVR	Linear kernel
MLP	One hidden layer with 50 neurons, 500 maximum iterations

Table 6.3

Descriptive statistics of the cleaned dataset.

Variable	N	Mean	Std. dev.	Min.	Max.
carbono_bruto4	133 119	20.79	35.09	0.00	420.50
carbono_bruto	123 563	10.09	26.33	0.00	511.31
c4	103 790	38.83	47.27	0.48	883.46
c	131 942	29.58	40.62	0.44	873.36

on ensemble performance to be analysed. On the stacked predictions of each configuration, the meta-models listed in Table 6.2 were trained.

By evaluating all combinations of base model configurations with the different meta-models, a set of ensembles is obtained that allow a systematic study of which subsets of base models are most complementary and which type of meta-model best exploits the information contained in their predictions.

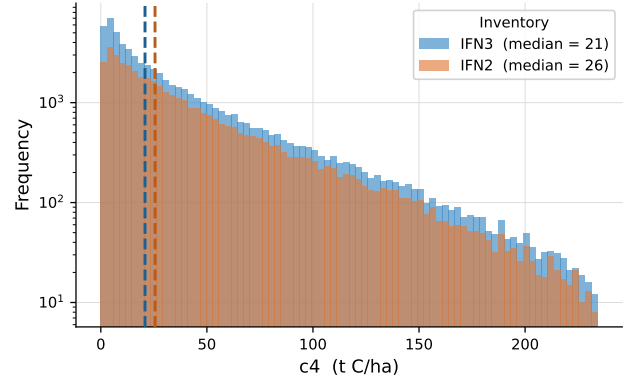
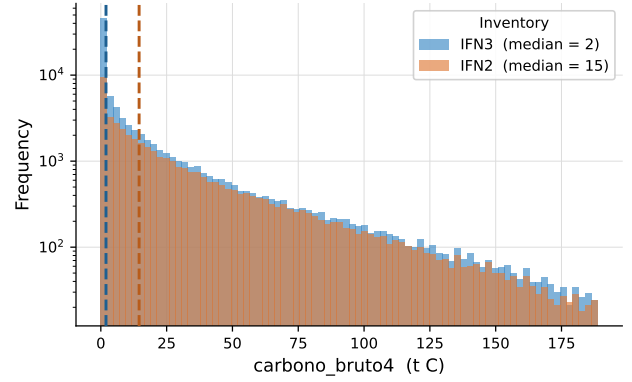
6.3. Final training data

After applying the eligibility and filtering criteria described in Section 5.4.1, the final dataset used for model fitting consists of:

- **IFN2:** Total plots = **88,696**
 - Cases with $c4 > c$: **41,761**
 - Cases with $\text{carbono_bruto4} > \text{carbono_bruto}$: **43,539**
- **IFN3:** Total plots = **171,157**
 - Cases with $f_{ccarb} \geq 20$: **164,364**
 - Cases with $f_{ccarb} \geq 20$ and $c4 > c$: **62,029**
 - Cases with $f_{ccarb} \geq 20$ and $\text{carbono_bruto4} > \text{carbono_bruto}$: **89,580**

Table 6.3 summarises the main descriptive statistics of the target variables used in the modelling. It can be observed that the variable *carbono_bruto4* has a mean of 20.79 and a standard deviation of 35.09, while the variable *c4* shows notably higher values (mean of 38.83 and standard deviation of 47.27). A histogram of the distribution of the target variables can be found in Figure 6.1.

On the other hand, Figure 6.2 shows the distributions of the target variables separated by inventory. It can be observed that the histogram of the *carbono_bruto4* variable

(a) Histogram of the *c4* variable on a logarithmic scale.(b) Histogram of the *carbono_bruto4* variable on a logarithmic scale.**Figure 6.1:** Histogram of the *c4* and *carbono_bruto4* variables in the cleaned dataset.

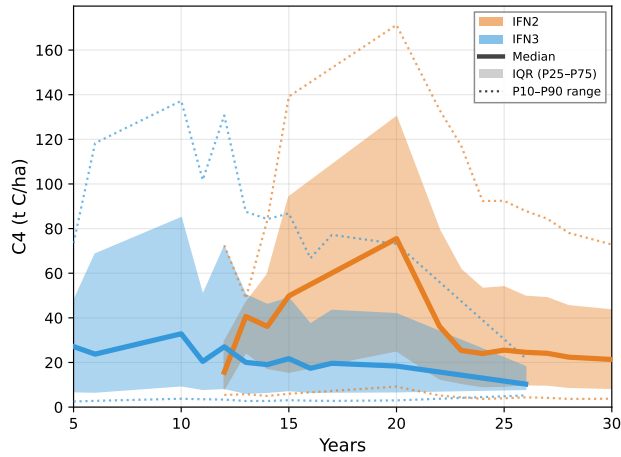
is notably more concentrated at low values than that corresponding to *c4*, for which the amount of carbon has been standardised per unit area.

6.3.1. Effect of the period on carbon

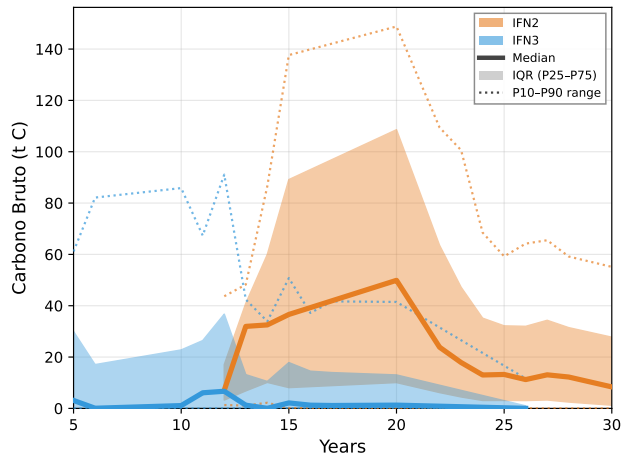
The influence of *periodo* on the carbon variables was evaluated using one-way ANOVA. The analyses carried out show that *periodo* has a significant effect on both variables. For *c4*, a test statistic of $F = 143.49$ ($p < 0.001$) was obtained, while for *carbono_bruto4* the value was $F = 161.08$ ($p < 0.001$). These results indicate that the differences observed between periods are not random, but reflect systematic variations associated with the sampling time, confirming that *periodo* constitutes a relevant explanatory factor in the dynamics of forest carbon.

7. Results

The following presents the results obtained on the test dataset, organised by source inventory (IFN2, IFN3 or IFN2+IFN3) and target variable (carbon in tC/ha or total tC). For each combination, the three best base models and the three best stacking ensemble models are shown, ordered by descending R^2 . The left column indicates the evaluation inventory, i.e., the inventory on which the predictions were



(a) Distribution of the *c4* variable across values of the *periodo_agrupado* variable.



(b) Distribution of the *carbono_bruto4* variable across values of the *periodo_agrupado* variable.

Figure 6.2: Distribution of the *c4* and *carbono_bruto4* variables in the cleaned dataset (after applying filters) separated by inventory.

made. When a model carries the subscript *global*, it indicates that it was trained with the combined IFN2+IFN3 data, even if it was evaluated on a single inventory. Models without a subscript were trained exclusively on the inventory on which they were evaluated.

The complete tables with all evaluated models are included in Appendix A.

Table 7.1

Top-3 base and *stacking* models by source inventory and target variable, ordered by descending R^2 . Models with the subscript *global* were trained with combined IFN2+IFN3 data.

Evaluation inventory	#	Model	Type	R^2	RMSE	MAE
<i>Target variable: c4 (tC/ha)</i>						
IFN2	1	LightGBM _{global}	Base	0.719	26.806	14.689
	2	LightGBM	Base	0.712	26.435	14.755
	3	CatBoost _{global}	Base	0.708	27.359	14.832
	1	Stack5 GradientBoosting	Stacking	0.724	25.889	14.531
	2	Stack1 MLP _{global}	Stacking	0.722	26.683	14.582
	3	Stack2 MLP _{global}	Stacking	0.721	26.726	14.475
IFN3	1	LightGBM	Base	0.858	17.816	9.246
	2	XGBoost	Base	0.851	18.232	9.203
	3	CatBoost	Base	0.849	18.385	9.746
	1	Stack4 GradientBoosting	Stacking	0.866	17.303	9.134
	2	Stack5 GradientBoosting	Stacking	0.865	17.369	9.037
	3	Stack2 GradientBoosting	Stacking	0.863	17.537	9.097
IFN2+IFN3	1	LightGBM _{global}	Base	0.787	22.767	11.650
	2	XGBoost _{global}	Base	0.784	22.952	11.590
	3	CatBoost _{global}	Base	0.783	22.990	11.607
	1	Stack5 MLP _{global}	Stacking	0.794	22.387	11.321
	2	Stack2 MLP _{global}	Stacking	0.794	22.405	11.403
	3	Stack4 MLP _{global}	Stacking	0.792	22.527	11.314
<i>Target variable: carbono_bruto4 (tC)</i>						
IFN2	1	CatBoost _{global}	Base	0.765	18.533	10.402
	2	LightGBM _{global}	Base	0.762	18.627	10.417
	3	XGBoost _{global}	Base	0.762	18.644	10.428
	1	Stack5 MLP _{global}	Stacking	0.768	18.405	10.210
	2	Stack5 LinearReg. _{global}	Stacking	0.768	18.418	10.306
	3	Stack4 MLP _{global}	Stacking	0.767	18.433	10.229
IFN3	1	LightGBM	Base	0.898	10.641	4.687
	2	CatBoost	Base	0.896	10.737	4.789
	3	XGBoost	Base	0.895	10.761	4.713
	1	Stack4 MLP	Stacking	0.902	10.407	4.517
	2	Stack5 GradientBoosting	Stacking	0.901	10.433	4.540
	3	Stack5 MLP	Stacking	0.901	10.447	4.454
IFN2+IFN3	1	CatBoost _{global}	Base	0.845	13.846	6.615
	2	LightGBM _{global}	Base	0.841	14.006	6.654
	3	XGBoost _{global}	Base	0.840	14.054	6.655
	1	Stack5 MLP _{global}	Stacking	0.847	13.756	6.399
	2	Stack4 MLP _{global}	Stacking	0.847	13.766	6.429
	3	Stack4 Lin.Reg. _{global}	Stacking	0.846	13.782	6.533

8. Discussion

First, the “global” models will be analysed, i.e., those trained on all data (IFN2 and IFN3) simultaneously. Subsequently, models trained on a single inventory will be analysed and compared with the global ones.

8.1. Global models

8.1.1. Variable *c4* (in tonnes of carbon per hectare)

Base models

The variable *periodo* (the number of years between the measurement and the prediction) is of great importance in the study of the models. For this reason, Figure 8.1a includes metrics as a function of this variable, showing the evolution of RMSE and %RMSE in quantiles as a function of the *periodo* variable, together with a histogram that allows visualisation of the amount of data in the training set for each value of *periodo*.

We can observe something we could anticipate: the difficulty of prediction depends on a combination of how far into the future the prediction is desired and the quality and quantity of training data. Starting in the IFN3 range, we can observe that for the first years (5, and 6), despite having few data points, errors are modest. As the period increases (between 10 and 17 years), mean and median errors remain relatively stable despite generally having more training data. Large extreme errors (quantile 90) experience a widespread increase. When we enter the range of years where training data come mainly from the IFN2 (the range between 20 and 30 years), although the median and the low-to-medium values (quantile 25) remain stable, the medium-to-high (quantile 75) and large (quantile 90) values increase substantially. This indicates that, even with a large amount of training data (especially for years 27 and 28), the combination of lower data quality (always comparing with IFN3) and prediction at distant values makes model prediction harder. Furthermore, as can be seen in Figure 6.2, the values for the first years of IFN2 show high variability, being mostly higher than those of IFN3 for the same years, with greater variability. This, combined with the fact that the amount of data for those years is not excessively large, causes the model to obtain worse predictions.

The fact that median values remain relatively stable across all periods indicates that the model correctly assimilates the behaviour of the target variable both with IFN2 and IFN3 data. On the other hand, the high quantile 90 errors can be explained by several factors: the high variability of the dataset, the smaller number of field variables collected in the IFN2 (which reduces available information) and the greater inherent difficulty of predicting over long time horizons. This combination makes particular or atypical cases less recognisable by the model, especially when they come from the IFN2, where discriminative capacity is lower than in the IFN3. The aforementioned variability is clearly visible in Figure 8.2a, which shows predicted values versus actual values for the model with the best metrics, along with a distribution histogram.

Stacking models

The incorporation of *stacking* schemes does not produce substantial, though systematic, increases in the coefficient of determination with respect to the best individual models. Looking at Table A.10 we can observe that stacking models present better metrics in the vast majority of cases, always compared with the base models. The best parameters are obtained by the stacking model with configuration 5, which retains all models with competitive performance, together with the neural network meta-model.

The fact that all ensembles improve upon the base models is a sign that this improvement is not a statistical exception, but a real improvement. That is, it is not the case that the metrics are better due to stochasticity in the meta-model parameters, but because the stacking genuinely improves the final result. Obviously, retraining the meta-models with other parameters would change the metrics, but the fact is that the function of improving predictions is indeed achieved with the ensembles. However, the improvement is small, which may mean that depending on the case, simplicity of using one of the base models is preferred over one of the ensembles.

Figure 8.1b shows the evolution of RMSE and percentage RMSE in quantiles for the ensemble model with the best metrics, the stacking with configuration 5 and MLP meta-model. We can observe that the behaviour is practically identical to that of LightGBM, which achieved the best metrics among the base models. On the other hand, in Figure 8.2b we can see the scatter plot of predictions versus actual values for the stacking model with configuration 5 and MLP meta-model for the variable *c4* (tC/ha).

Regarding the structure of the ensembles, the best results are obtained when high-quality, similar-nature base models are combined (mainly *gradient boosting* variants) and meta-models of moderate complexity, such as MLP or linear SVR, are used. In contrast, stacks with few base models or those incorporating excessively flexible meta-models, such as Random Forest at the second level, tend to offer lower performance, probably due to the low dimensionality of the meta-predictor space or unnecessary overfitting of residual noise.

8.1.2. Variable *carbono_bruto4* (in tonnes of carbon)

Base models

In the same way as in the previous sections, Figure 8.1c shows the evolution of RMSE and %RMSE in quantiles as a function of the *periodo* variable for the CatBoost model (the one that achieved the best metrics among the base models) with IFN2 and IFN3 as explanatory variables for the variable in tonnes of carbon, together with the histogram of the distribution of training data as a function of the period variable.

Referring to Figure 8.1c we observe the same qualitative behaviour as in the case of the *c4* variable (with the particularity that units and value range change): the error between quantiles 25 and 75 remains at relatively small

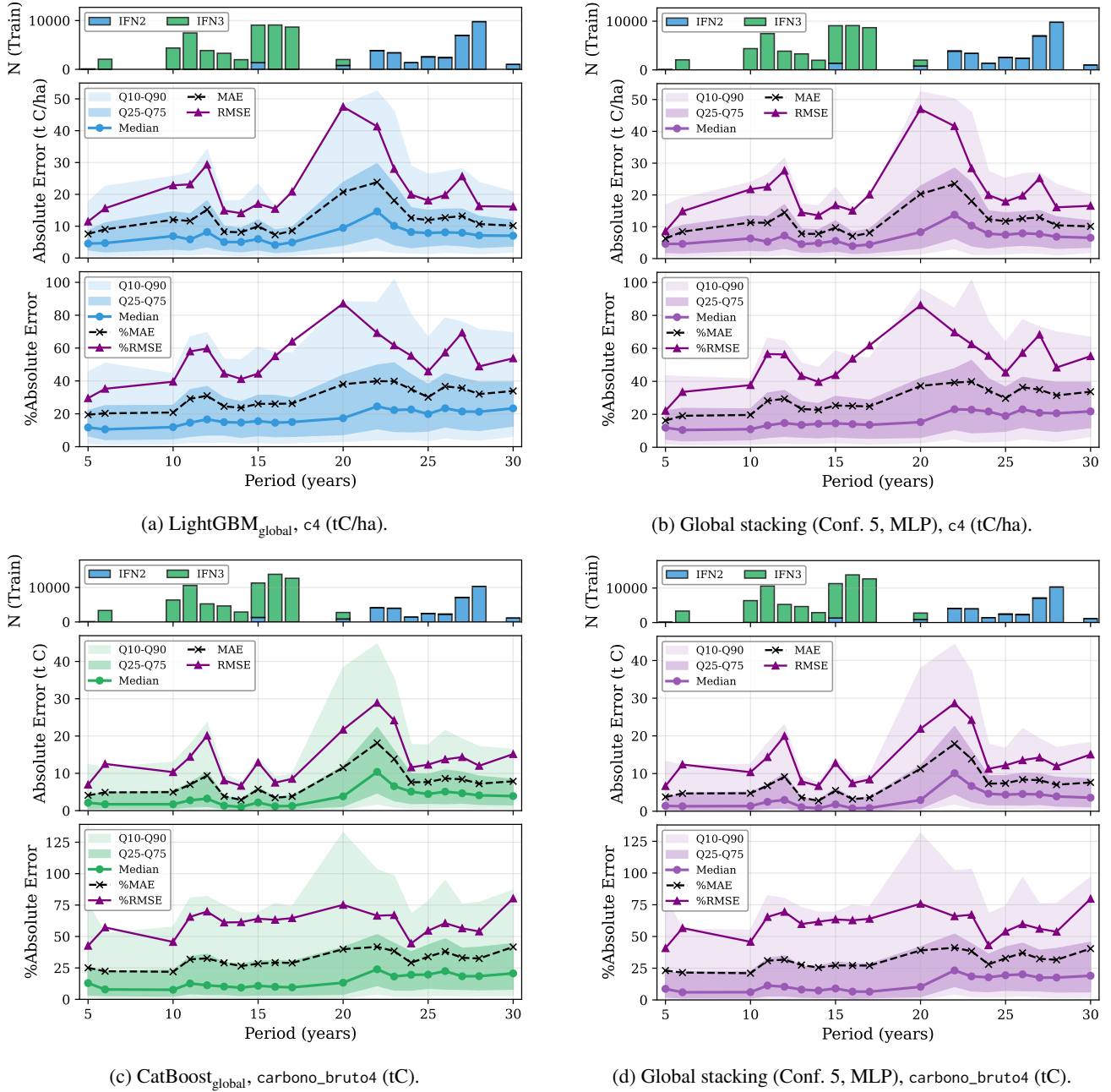
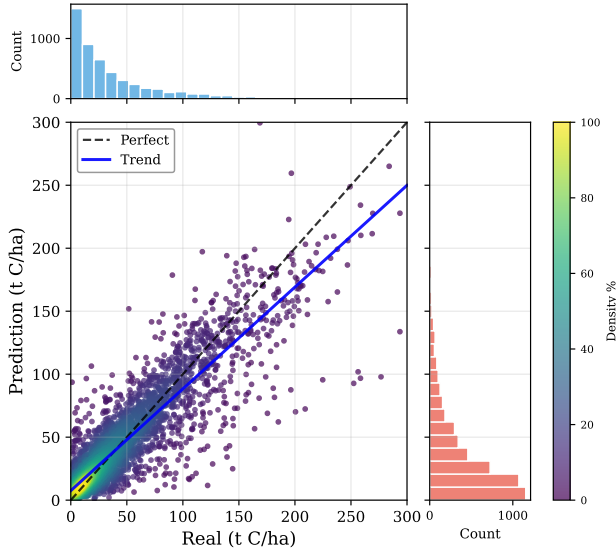
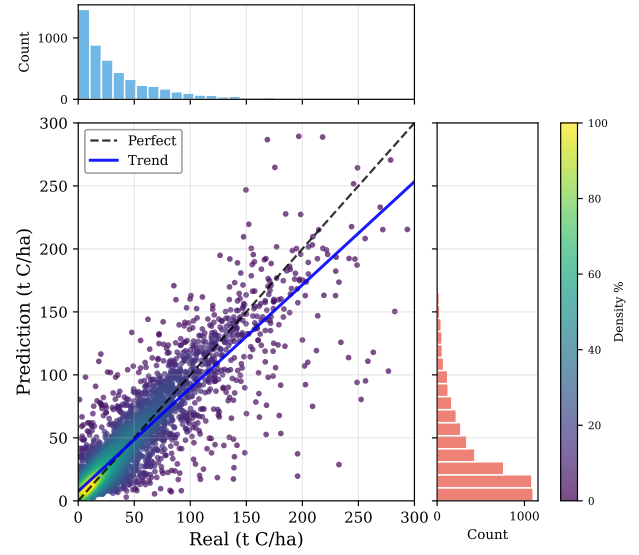


Figure 8.1: Evolution of absolute error and percentage absolute error in quantiles as a function of the period for the best global models (IFN2+IFN3), base and *stacking*, for each target variable.

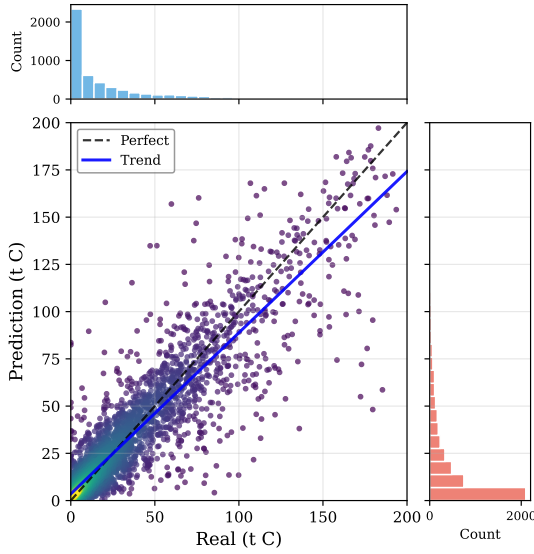
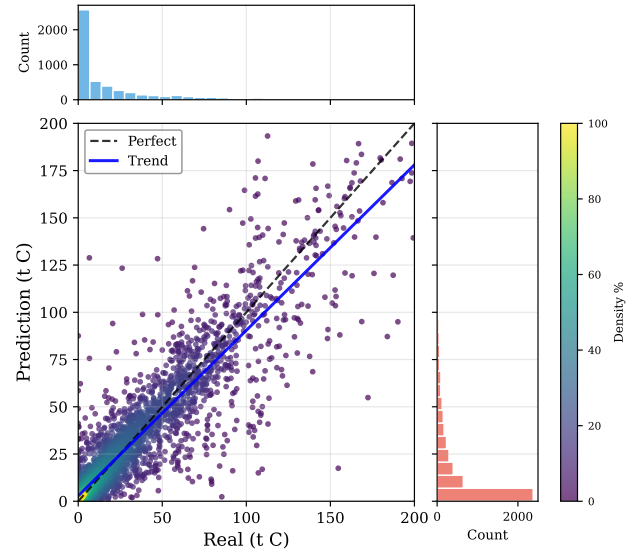
and constant values, although it is true that it starts smaller and gradually increases as the period variable increases. Extreme errors (quantile 90) remain high. In fact, looking at the highest values of the percentage RMSE for quantile 90, we observe that notably higher values are reached than in the case of the *c4* variable, reaching up to 130% when for *c4* it barely exceeded 100% for the worst case. This is noteworthy, since the global metrics for predictions of the variable *carbono_bruto4* (see Table A.11) are systematically better than those of the variable *c4* (see Table A.9). That is, the variable *carbono_bruto4* (tonnes) is generally easier for models to predict than the variable *c4* (tonnes per hectare),

but the opposite occurs in particular cases, where the highest errors shoot up in a more exaggerated manner than in the same case for the *c4* variable. The increase in error when moving from the IFN2 to IFN3 year range is also observed, caused by the increase in variability of the second inventory data (see Figure 6.2).

As in the previous sections, Figure 8.2c shows a scatter plot of predictions versus actual values for the CatBoost model for the case under consideration.


 (a) LightGBM_{global}, c4 (tC/ha).


(b) Global stacking (Conf. 5, MLP), c4 (tC/ha).


 (c) CatBoost_{global}, carbono_bruto4 (tC).


(d) Global stacking (Conf. 5, MLP), carbono_bruto4 (tC).

Figure 8.2: Prediction density versus actual values for the best global models (IFN2+IFN3), base and *stacking*, for each target variable.

Stacking models

The results obtained in this section are very similar to those obtained with the stacking models for the c4 (tC/ha) variable. That is, we find a systematic improvement in all stacking models over the base models, but that improvement is small. This causes the results to be strictly better, as can be seen in Table A.12 compared with the base model metrics in Table A.11. Again, the fact that the improvement of stacking models over base models is systematic indicates that we are not facing a statistical incident, but rather that the stacking format is capable of improving predictions by choosing the best decisions from each model. However, as noted above, the complexity of training five models in addition to the

meta-model may be inconvenient compared to the possibility of using, for example, the best of the individual models.

The ensemble model that provides the best metrics for predicting the carbono_bruto4 (tC) variable is configuration 5 with an MLP meta-model, as was also the case for the c4 (tC/ha) variable. The evolution of RMSE and percentage RMSE in quantiles as a function of the periodo variable for this model can be visualised in Figure 8.1d.

On the other hand, Figure 8.2d shows a scatter plot of predictions versus actual values for the stacking model with configuration 5, MLP meta-model and the variable carbono_bruto4 (tC).

8.2. Individual models trained on IFN2 or IFN3 only.

When comparing the performance of “global” models (trained on the combined IFN2 and IFN3 dataset) versus “local” models (trained exclusively on IFN2 or IFN3), differentiated behaviour is observed depending on the test inventory. For IFN3, local models outperform global ones in terms of accuracy metrics. Conversely, for IFN2, global models show superior performance.

A possible explanation for this phenomenon lies in the quality and quantity of information contained in each inventory. The IFN2 data, being less informative or having fewer variables, appear to benefit from joint learning with IFN3 data, allowing models to capture additional patterns and improve their generalisation capacity on the second inventory. In contrast, for IFN3, whose data possess higher intrinsic quality, the inclusion of IFN2 records during training could be introducing noise or unwanted variability, which ultimately penalises the accuracy of the final model predictions on this dataset.

Figure 8.3a shows a comparison between the best global vs. local base models (top), best global vs. local stacking models (middle) and best base model vs. best stacking model (bottom). We observe that in general the lower errors for IFN2 are committed by the global models, while for IFN3 it is the local models that do the better job.

8.2.1. Assimilation of target variable behaviour

With the metrics we have set out and the analysis carried out in this section, it seems clear that, while the results are far from perfect, the models are capable of understanding the logic and meaning of each variable to obtain a prediction with a degree of accuracy that depends on the nature of the instance in question. Another test we can perform to check whether the model results are logical is the one shown in Figures 8.4 and 8.5. These figures show the evolution of several particular cases as time progresses between measurement and prediction. The selected cases are real within the available data. They were selected with the aim of showing how the models behave for more or less common initial carbon configurations. To this end, for each data instance, the `npies_x` variables were summed, and then the values closest to percentiles 10, 35, 65 and 90 for the resulting variable were selected. This avoids potential problems arising from selecting unrealistic examples.

We can highlight the following features:

- The lines of the base models and ensembles for each case are very close together, which is consistent with the obtained metrics, which were very similar to each other.
- The general trend is upward, i.e., the models correctly capture the behaviour of tree growth over time for the studied years.
- In year 17 there is a widespread abrupt increase, which is more exaggerated the smaller the percentile of the case study. This is easy to understand by looking at the data, since the values of the variable to be predicted

experience a generalised increase when moving from the IFN2 years to those of the IFN3, as can be seen in Figure 6.2. This change (where the model begins to ingest IFN2 data instead of IFN3 data) occurs mainly in year 17. This can be seen in the histogram of Figure 8.1d, for example. The reason why the jump occurs at small percentiles is that these small values in the IFN2 data barely exist or are less common. The model has seen that for those years carbon values are, in general, higher, and that is what it predicts. In fact, Figure 6.2 shows that for the initial years of the IFN2 data, carbon values increase and then decrease, and this is exactly what we see in the predictions: an increase followed by a decrease.

- We can also see an abrupt increase in many cases when moving from year 29 to 30, especially in the case of the variable in tonnes of raw carbon (Figure 8.5). These years coincide with a scarcity of data, as can be seen in the histogram of Figure 6.2. It is to be expected that predictions for those years will be less reliable, especially for an extreme value.

8.2.2. Model performance as a function of the target variable value

If we analyse the global metrics in the tables of Section 7, we notice that RMSE values are notably larger than MAE values, specifically around twice as large in most cases. Because RMSE penalises large errors much more heavily than MAE, this is indicative of the presence of outliers in the error. We have already seen throughout this section that the largest errors occur in the range of years in which the IFN2 is the main data source, as can be seen in Figures 8.1a, 8.1b, 8.1c and 8.1d, and the main conclusion is that the quality or quantity of information contained in this inventory is inferior. However, this does not fully answer the question of in which cases the model predicts best, beyond the obvious fact that prediction will be better the closer in time and the more data available for that year. This is why Figures 8.6 and 8.7 can be useful. These figures show the distribution of errors, specifically RMSE and SMAPE, as a function of the target variable for `c4` and `carbono_bruto4` respectively, in deciles for the base models.

We quickly notice the following: predictions for small values of the target carbon (first deciles) are poor (large SMAPE) despite having the smallest absolute error (small RMSE). Then, as the target carbon increases, RMSE increases slightly while SMAPE decreases much more rapidly, reaching values close to 25%. In both cases the best results are achieved when the target carbon is larger, despite a significant increase in RMSE between the second-to-last and last decile. Since this occurs for all models, it suggests that prediction difficulty is not homogeneous across all situations or amounts of carbon, and that despite the lowest-carbon percentiles having the smallest RMSE, model accuracy is not sufficient to make a decent prediction in situations with little biomass. This was predictable, since a larger absolute error

Forest carbon stock estimation in Spain

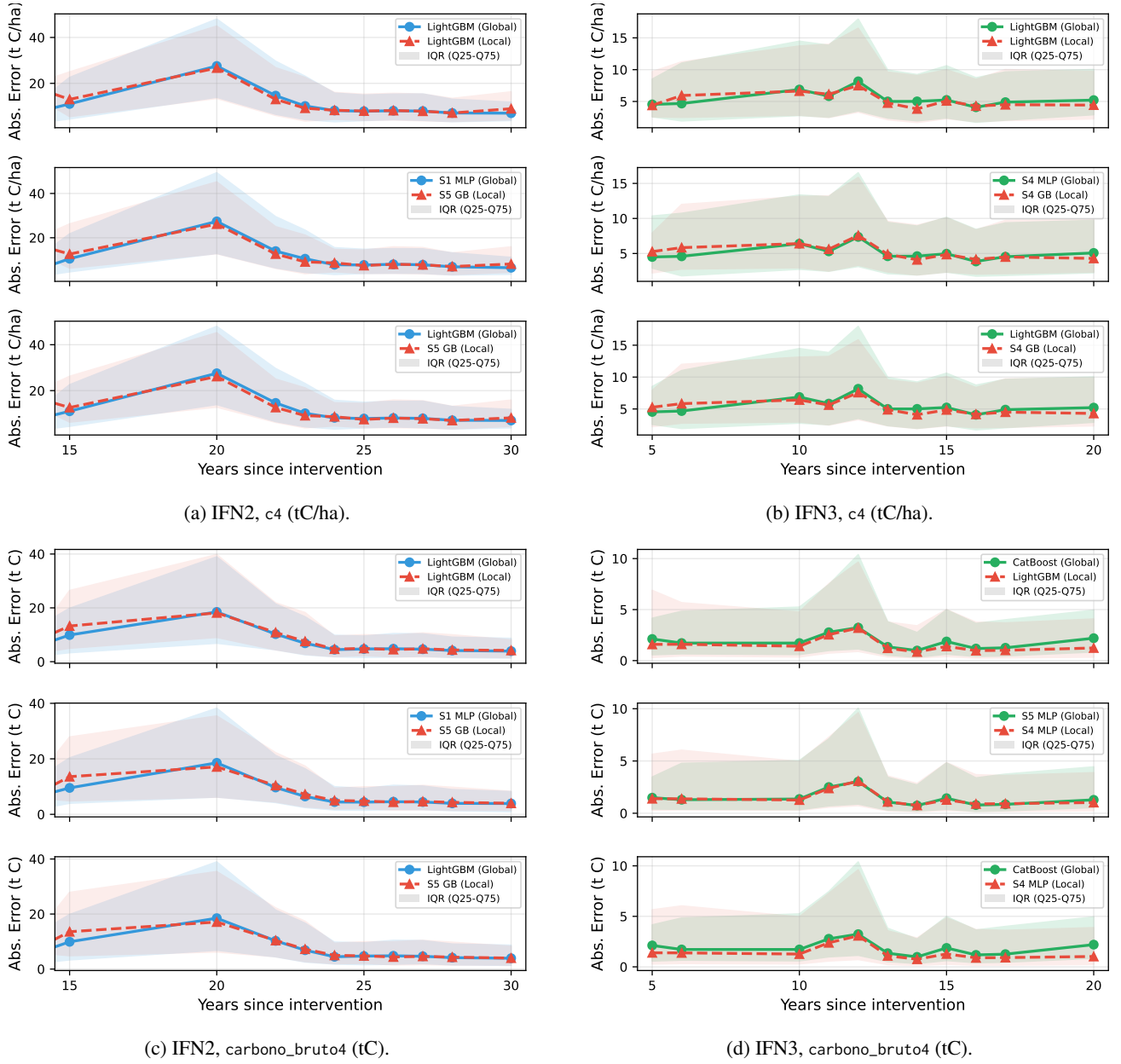


Figure 8.3: Comparison of absolute error by inventory and target variable: best global vs. local base models, global vs. local stacking models, and best base model vs. best stacking model.

for a larger forest plot does not necessarily imply a worse relative error.

Also noteworthy is the fact that the initial deciles have very close carbon ranges, especially for the `carbono_bruto4` variable, where the first decile ranges from 0.0003 tC to 0.0020 tC. This large number of values with such small carbon does not help models obtain a good prediction. The higher densities at small carbon values are clearly visible in Figures 8.2a, 8.2b, 8.2c and 8.2d.

8.3. Model interpretability: SHAP analysis

Thanks to the SHAP value analysis we can add explainability to the models, finding out which variables the model gives more importance to and in which cases these values

cause the carbon prediction to be higher or lower. In Table 8.1 we can observe the mean importance rank of variables with the highest absolute SHAP value, both averaged across all cases (first column “Mean”) and by source inventory and target variable. In this table we see that the most consistently important variables are `npies`, `especie_id`, `ocupa` and `radio`, i.e., variables purely related to the forest structure. The rest of the variables in the table have more variability across the different configurations, where meteorological variables (which appear to be more important than the climatic variable we added, the Martonne index) and edaphic variables are mainly interleaved.

The meteorological variable that consistently has the most importance according to our analysis is the mean value

Table 8.1

Mean SHAP rank per variable across LightGBM, XGBoost, CatBoost, GBDT and MLP (n=2000 samples; rank 1 = most important).

Variable	Mean	IFN2 c4	IFN2 carbono_b4	IFN3 c4	IFN3 carbono_b4	IFN2+3 c4	IFN2+3 carbono_b4
npies* (Density)	1	1	1	1	1	1	1
especie_id*	2.2	2	2	2.2	2.8	2.2	2.2
ocupa	2.9	3.2	3	3	2.2	2.8	3
radio	4.2	4.2	4.2	4.2	4.2	4.2	4.4
gndvi_mean_verano	6.5	5.6	5	7	7	8	6.6
tipo_especie*	7.1	11.6	8.8	4.6	5.8	6	6
periodo_agrupado	8.4	13	8	10.6	8.6	5.2	5
skt_mean_verano	9.6	6.8	—	10.4	—	11.6	—
rocosidad_id*	11	9	7.6	17.2	12	10.2	9.8
fccarb	11.4	—	—	9	10	14	12.6
evi_mean_primavera	11.8	12	10.6	13.6	11.2	13.2	10.4
elevacion	12.6	12.6	7.4	15.8	12.8	14.4	12.4
tratmasa_id*	13.2	—	—	8	7.2	18.8	18.8
estado_id*	13.5	21.8	12.8	17.2	12.6	8.8	8
ndii_mean_primavera	14.5	17	11.6	14.2	15.6	15.4	13.4
skt_mean_primavera	15.1	17.4	—	13.4	—	14.6	—
pr_sum_otoño	15.7	16.4	—	14.4	—	16.4	—
pr_sum_primavera	15.7	10.6	—	17.6	—	19	—
pr_sum_invierno	16	8.6	—	21.8	—	17.6	—
martonneidx_id*	16.4	18.2	10.2	24.2	13.4	20.6	11.8
skt_std_verano	18.4	13.2	—	22.8	—	19.2	—
skt_std_primavera	18.4	15.8	—	20	—	19.4	—
orientacion_cat*	18.9	19.8	14.4	23	17	21.4	18
modcomb_id*	18.9	—	—	18.8	14	26	16.8
gndvi_std_primavera	19.8	20.8	14.6	24	18	24.8	16.6
pendiente_cat*	20	20.4	15.4	26.2	18	23.8	16.4
pr_sum_verano	20.3	19.4	—	21.2	—	20.4	—
nivel2_id*	20.6	—	—	20.6	17.2	27	17.6
disesp_id*	23.3	24.6	16.4	29	20.4	29	20.2

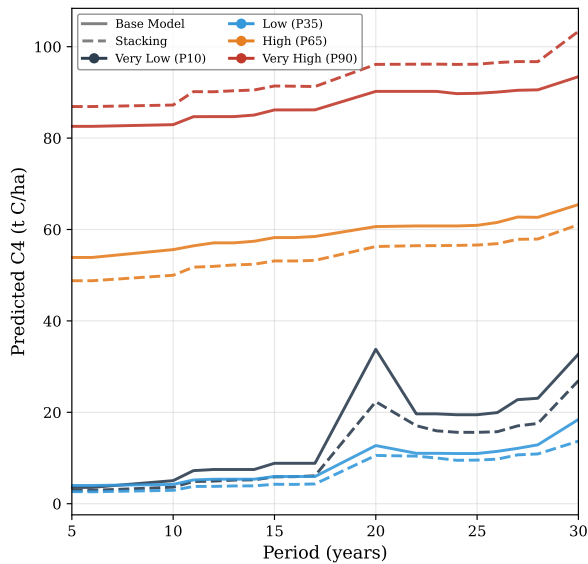


Figure 8.4: Evolution of the amount of carbon in tonnes per hectare (c4) for several scenarios.

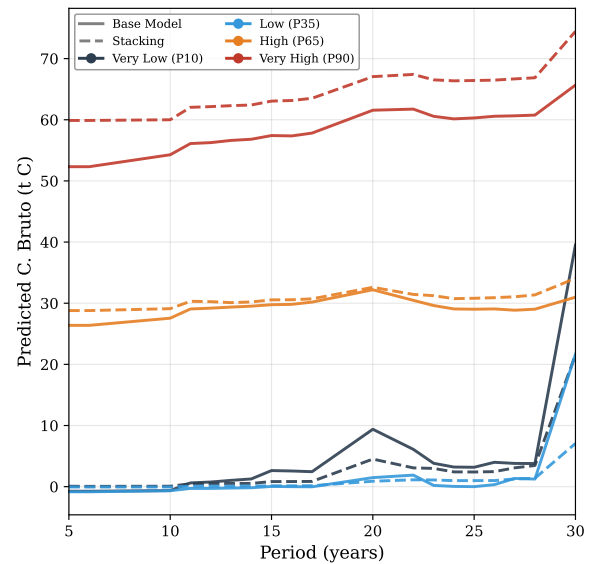


Figure 8.5: Evolution of the amount of carbon in tonnes (carbono_bruto4) for several scenarios.

of the GNDVI index in summer (gndvi_mean_verano), which stands out both by position and consistency. This is expected,

since it is in summer when forest vegetation has the greatest vigour, and the GNDVI index is a more useful indicator in

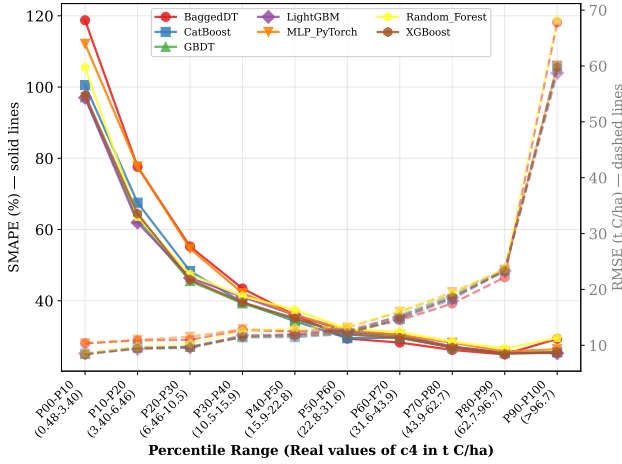


Figure 8.6: Distribution of SMAPE and RMSE as a function of the target variable for c_4 in deciles.

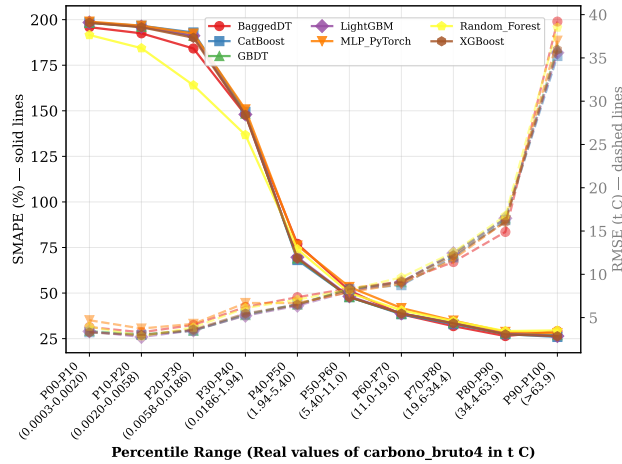


Figure 8.7: Distribution of SMAPE and RMSE as a function of the target variable for carbon_bruto4 in deciles.

this season compared to others such as NDVI, as it does not have saturation problems in dense canopies [33, 19]. Furthermore, this index serves to estimate photosynthetic activity, a variable directly related to both the amount and health of vegetation.

In Figures 8.8, 8.9 and 8.10 we can see a comparison between the SHAP variables of the different configurations for the LightGBM model and the target variable c_4 . It should be noted that for the two most important variables for the analysis (npies , especie_id and ocupa), the x-axis scale is different from the rest, as their SHAP value is notably larger. This was done to facilitate visualisation. The variables whose text is coloured are categorical variables (also marked with an asterisk, *), and their colour indicates the treatment applied for inclusion in the figure. First, there is npies . This variable actually consists of several, one for each diameter class (npies_5 , npies_{10} , npies_{15} , etc.). The green text colour indicates that for each diameter class the analysis was performed individually and then all classes were combined in the same row of the plot. That is, when

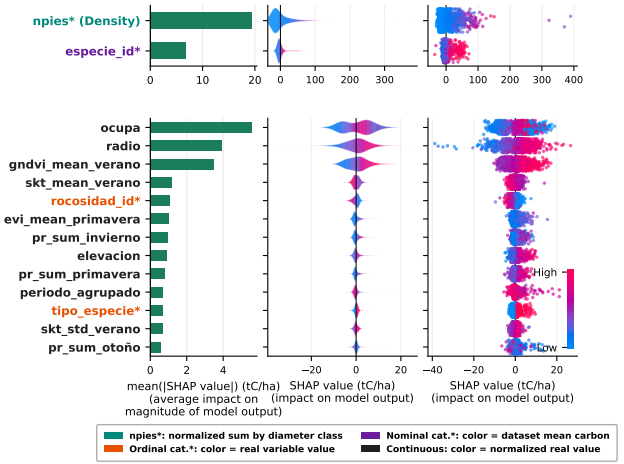


Figure 8.8: SHAP analysis for the LightGBM model trained on IFN2 data with target variable c_4 (tC/ha).

the colour of the point indicates that the variable value is “high”, this is normalised to each diameter class, since a high value of the variable npies_5 is not comparable with a high value of npies_{70} . The purple text colour indicates that the categorical variable is purely nominal, i.e., it has no order despite being passed to the model encoded in numerical format. This occurs with the variable especie_id , where a higher numerical value does not indicate, for example, a higher carbon absorption. In this case, the colour of each point expresses the mean of the target variable for each species, so that all points of the same species will have the same colour. This often produces an obvious result: species that absorb more carbon will be those with a higher SHAP value, and therefore will be located generally to the right (which is indeed the case). However, it is the only way found to include this variable in the figure, since it had to be present due to its high importance for the model. If the text colour is orange, it means the categorical variable is ordinal. That is, the category follows an order that makes sense to represent by colours. For example, the variable rocosidad_id goes from value 1, “no stoniness”, to 5, “rocky”, with the increase in the variable related to an increase in the indicated characteristic. Finally, black text colour indicates that the variable is continuous and is treated as expected.

In Figures 8.8, 8.9 and 8.10 we see that the two most important variables are always npies and especie_id , as we had seen in Table 8.1. The remaining variables change order depending on the case, but speaking of the whole set, the variables identified as important tend to be common across the three cases in the Figures. It should be noted that, although the positions of the variables do not exactly match between the case of the model trained solely with IFN2 data (Figure 8.8) and the model trained with IFN2 and IFN3 data (Figure 8.10), the variables present in the top 15 shown are practically the same in total (except for tratmasa_id and estado_id which are swapped). Variables exclusive to IFN3, such as tratmasa_id , also appear, indicating that these new variables incorporated into successive inventories provide

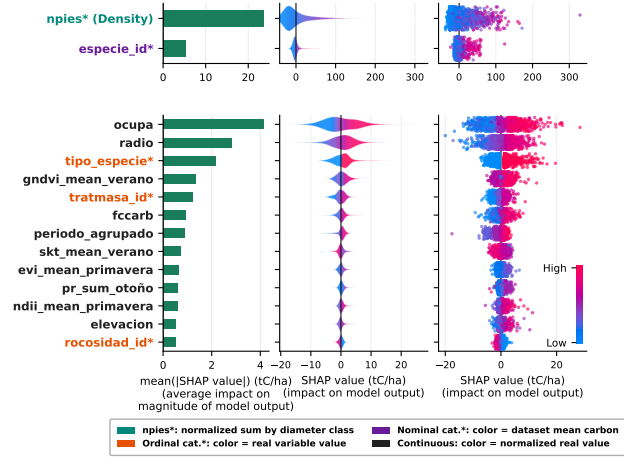


Figure 8.9: SHAP analysis for the LightGBM model trained on IFN3 data with target variable c_4 (tC/ha).

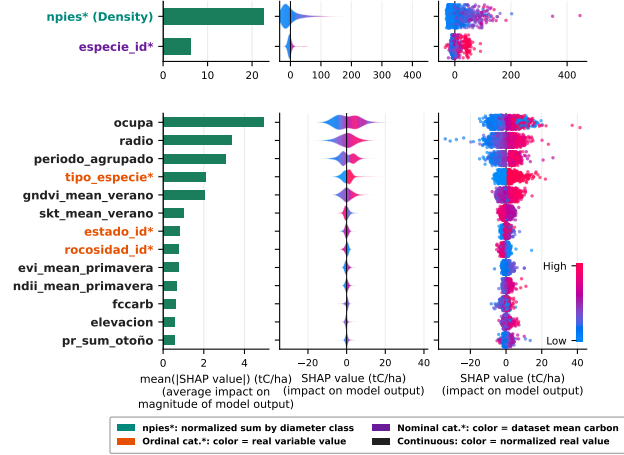


Figure 8.10: SHAP analysis for the LightGBM model trained on IFN2 and IFN3 data with target variable c_4 (tC/ha).

useful information for the study of forests, and specifically for studying carbon capture.

Looking at the values in Figures 8.8, 8.9 and 8.10 we can highlight some interesting results that support the idea that the model has correctly incorporated the information:

- Looking at the variable `npies`, we observe that a higher number of initial trees corresponds to a higher SHAP value, i.e., more carbon.
- As stated earlier, for the variable `gndvi_mean_verano` a higher SHAP index is obtained for cases where its value is also higher. This is logical, since the GNDVI index measures photosynthetic activity. A higher value of this index corresponds to a greater presence of active vegetation, and therefore greater carbon capture.
- The variable `periodo_agrupado` indicates the number of years between the measurement and the moment at which we wish to make the estimate, so it is expected that a higher value corresponds to a higher SHAP

value, i.e., more carbon. This is what we find for the IFN3 case (Figure 8.9) and IFN2 and 3 (Figure 8.10). Curiously, for the IFN2 case (Figure 8.8) this relationship is less clear, which also coincides with the SHAP model assigning it relatively lower importance.

- The variable `tipo_especie` has a very clear distinction in all cases. This is a binary variable in which value 0 corresponds to conifers and value 1 corresponds to broadleaves. According to the analysis, broadleaf species obtain on average a higher SHAP index, i.e., a greater amount of carbon associated with them. This is consistent with the literature, and specifically with studies carried out using the Spanish National Forest Inventory, which quantify the mean carbon *stock* in broadleaf stands at 48.5 ± 0.3 Mg/ha compared to 41.8 ± 0.2 Mg/ha in conifers for mainland Spain [34]. For further detail, Figure 8.11 presents a breakdown of the SHAP index by species for the LightGBM model trained on data from both inventories and with c_4 as the target variable. It can be observed that broadleaves occupy almost all the positions in which carbon increases, with *Eucalyptus globulus* as the first species, a species that has been causing controversy in northern Spain for years due to its possible characterisation as an invasive species [35, 36]. However, it should be noted that this result reflects the biomass inventoried in the IFN, and does not imply that eucalyptus is the most appropriate option from a sustainable forest management perspective, given its high water consumption, its association with a higher fire risk, the loss of biodiversity it can cause, the difficulty of eradication, or the danger associated with the involuntary introduction of associated species, such as insects or fungi, displacing other native species [35, 37, 38].

In summary, SHAP analysis not only allows identifying which variables are most relevant for carbon prediction and quantifying their relative importance, but also constitutes an ecological validation tool for the model. The results obtained are consistent with knowledge available in the literature: variables such as tree density, photosynthetic activity, species type or time elapsed between inventories show the expected direction in their SHAP values, indicating that the model has correctly incorporated the input information and that its predictions respond to real ecological processes.

9. Conclusions

The main objective of this work is the development of artificial intelligence models capable of predicting the carbon that a forest plot will capture over a given time horizon, based on structural, edaphic, climatic and spectral information collected in the National Forest Inventories and

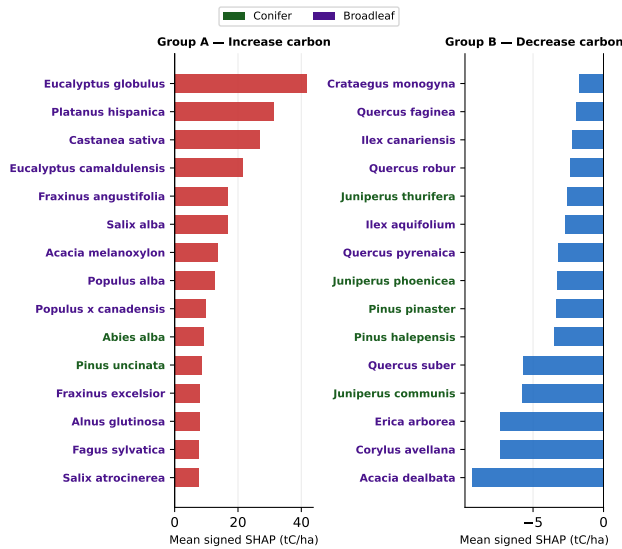


Figure 8.11: SHAP analysis for the LightGBM model trained on IFN2 and IFN3 data with target variable *c4* (tC/ha), focusing on species. A distinction is made between conifer and broadleaf.

auxiliary data sources. The results obtained allow us to affirm that this objective has been satisfactorily achieved.

Feature selection and preprocessing

The manual feature selection process (Section 6.1) allowed reducing an initial set of 445 candidate predictors to a compact subset of 44 variables, maintaining a balanced representation of all relevant ecological domains: stand structure, species composition, edaphic properties, terrain, climatology and spectral vegetation indices. This reduction not only facilitated the training of more efficient models, but also guaranteed the ecological interpretability of the selected predictors.

Model performance

Models trained with IFN3 data as explanatory variables generally achieve the best metrics (see Table 7.1). This is explained by the higher quality, methodological homogeneity and richness of variables collected in this inventory compared to the IFN2.

Regarding the target variable, the prediction of carbon in absolute tonnes (*carbono_bruto4*) is systematically more accurate in terms of global metrics than the prediction of carbon per unit area (*c4*, tC/ha), as can be seen by comparing Table 7.1. However, the decile analysis (Figures 8.6 and 8.7) reveals that for instances with very low carbon values (early biomass), relative errors (SMAPE) are high for both variables regardless of the model used. Models offer their best results in the intermediate-to-high biomass range, where the balance between data quantity, intrinsic variability and the magnitude of the target variable is most favourable.

The analysis of the *periodo* variable (Figures 8.1a, 8.1b, 8.1c and 8.1d) shows that error grows with the prediction time horizon, especially when training data come from the

IFN2 (periods exceeding 17 years). Nevertheless, median errors (quantile 50) remain relatively stable across all periods, indicating that the models correctly assimilate the average behaviour of forest growth even for long horizons. Extreme errors (quantile 90) are the most sensitive to data quality and temporal distance.

The sensitivity analysis on representative scenarios (Figures 8.4 and 8.5) confirms that the models qualitatively correctly capture the growth dynamics: the general trend is upward with the period, and the models reproduce the effects of the inventory changes (from IFN2 to IFN3) observed in the real data.

Global versus local models

The comparison between “global” models (trained on IFN2 and IFN3 jointly) and “local” models (trained on a single inventory), shown in Figures 8.3a–8.3d and in Table 7.1, reveals differentiated behaviour depending on the validation inventory. For IFN3, local models outperform global ones, since the inclusion of IFN2 data introduces heterogeneity that penalises prediction on higher-quality data. For IFN2, in contrast, global models are superior, as they benefit from the additional patterns learned with the IFN3 data. This result underscores that the choice of training set must be adapted to the time horizon and the target inventory.

Stacking models

The incorporation of stacking schemes produces systematic but modest improvements over the best base models (see Table 7.1). The stacking ensemble with configuration 5 and MLP meta-model is consistently the one that achieves the best metrics for both *c4* and *carbono_bruto4*. The fact that the improvement is systematic in all cases rules out it being a statistical artefact, but its limited magnitude raises the question of whether the additional complexity of stacking is justified compared to using the best individual model in a hypothetical production system. Gradient boosting-based algorithms, especially CatBoost and LightGBM, offer the best performance as base models in all evaluated scenarios.

Interpretability

The SHAP value analysis (Section 8.3, Table 8.1) confirms that structural stand variables—tree density (*npies*), species (*especie_id*), canopy cover fraction (*ocupa*) and plot radius (*radio*)—are the dominant predictors regardless of the source inventory or target variable considered. Among the auxiliary variables, the summer GNDVI index (*gndvi_mean_verano*) emerges as the most informative spectral predictor, consistent with its capacity to estimate photosynthetic activity without saturation in dense canopies. The direction of SHAP values is in all cases consistent with available ecological knowledge: higher tree density, greater photosynthetic activity and longer time horizons lead to higher predicted carbon; broadleaves systematically outperform conifers, in line with the mean *stocks* reported in the literature for mainland Spain [34]. This indicates that the models are not only numerically accurate, but have learned

relationships with genuine ecological meaning, which reinforces their interpretability and confidence for a possible practical application.

Comparison with the literature

In similar works, such as that of Fasihi et al. [26] for northern Italy, forest carbon values are predicted from remote sensing observations without the use of field-measured in situ data. Although the context is different—they predict the carbon present at the time of measurement, without temporal projection—the errors they report are notably higher than those obtained in this work, especially in the shortest prediction range (around 5 years), which is the most comparable. This highlights the added value of incorporating structural field information (IFN) compared to approaches based exclusively on remote sensing.

Limitations

The main limitation of the study lies in the heterogeneity and irregularity of the available data. The distribution of instances as a function of the *periodo* variable is very unequal: some years have thousands of samples while others have barely a few dozen. This irregularity, inherent to the nature of the Forest Inventories (conducted over different time periods and with methodologies that evolve between iterations), limits the model's capacity for certain time horizons, particularly in the range of years with fewer data and where the main source is the IFN2. Despite this, the models demonstrate reasonable generalisation capacity even under these conditions.

Future work and applicability

For a possible production application as a decision-support system for carbon project recommendations, it would be necessary to evaluate the extent to which IFN plots are representative of forest plantations intended for carbon credits, which may differ in density, management and composition. A natural extension of the model would be to incorporate LiDAR data, whose capacity to characterise the three-dimensional structure of the forest canopy has been shown in other studies to significantly improve biomass and carbon predictions [15]. Likewise, adapting the model to other geographical contexts or integrating it with IFN4 data as the target variable constitute promising lines of future work.

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Table A.1

Base Model Comparison for IFN2 (c4)

Model	R^2_{test}	RMSE _{test} (tC/ha)	MAE _{test} (tC/ha)	wMAPE (%)	SMAPE (%)	Bias (tC/ha)
LightGBM_{global}	0.719	26.806	14.689	35.364	47.104	-0.021
LightGBM	0.712	26.435	14.755	35.823	47.593	-0.401
CatBoost _{global}	0.708	27.359	14.832	35.710	47.895	-0.135
XGBoost _{global}	0.708	27.361	14.620	35.198	46.910	-0.171
GBDT _{global}	0.705	27.504	14.772	35.564	46.941	-0.142
XGBoost	0.704	26.779	14.773	35.867	47.452	-0.434
CatBoost	0.701	26.927	15.066	36.578	49.284	-0.753
MLP _{global}	0.686	28.335	15.668	37.722	50.547	0.280
BaggedDT	0.676	28.036	15.683	38.075	49.869	-0.195
GBDT	0.673	28.161	15.614	37.907	50.513	-0.699
RandomForest	0.663	28.595	15.807	38.378	49.750	-0.376
BaggedDT _{global}	0.658	29.574	15.933	38.361	50.974	-0.599
RandomForest _{global}	0.653	29.803	15.577	37.502	48.386	-1.499
MLP	0.637	29.650	15.543	37.735	49.907	-7.176
BayesianNN	0.606	30.904	17.384	42.205	56.359	-0.697
BayesianNN _{global}	0.588	32.482	17.725	42.675	56.630	-0.732
SVR _{global}	0.465	37.013	16.674	40.145	49.745	-8.320
SVR	0.445	36.667	17.361	42.150	52.112	-8.338

A. Results

The following presents the results obtained from several model evaluation metrics on the test dataset. The section is divided according to the source of the training data (IFN2, IFN3 or IFN2 and IFN3) and according to the variable to be predicted (tonnes of carbon or tonnes of carbon per hectare).

A.1. IFN2 as explanatory variable for c4 (tC/ha)

A.1.1. Base models

Once the models are trained, global parameters such as R^2 , RMSE and MAE are presented in Table A.1.

A.1.2. Stacking models

Table A.2 shows the performance summary of the stacking models for predicting the carbon variable in tC/ha using the dataset that employs IFN2 as the explanatory inventory.

Table A.2
Stacking Model Comparison for IFN2 (c4)

Model	R^2_{test}	RMSE _{test} (tC/ha)	MAE _{test} (tC/ha)	wMAPE (%)	SMAPE (%)	Bias (tC/ha)
Stack5 GradientBoosting	0.724	25.889	14.531	35.280	47.157	-0.579
Stack1 MLP _{global}	0.722	26.683	14.582	35.106	46.515	0.117
Stack2 MLP _{global}	0.721	26.726	14.475	34.848	46.320	-0.131
Stack5 MLP _{global}	0.721	26.742	14.502	34.913	46.347	0.077
Stack3 GradientBoosting	0.720	26.054	14.847	36.045	48.023	-0.747
Stack1 LinearRegression _{global}	0.719	26.827	14.609	35.173	46.712	-0.140
Stack1 Ridge _{global}	0.719	26.829	14.609	35.172	46.709	-0.142
Stack5 MLP	0.718	26.127	14.452	35.088	46.430	-0.607
Stack4 MLP _{global}	0.718	26.852	14.479	34.859	46.260	-0.328
Stack2 LinearRegression _{global}	0.718	26.877	14.505	34.922	46.440	-0.112
Stack2 Ridge _{global}	0.718	26.883	14.507	34.927	46.440	-0.114
Stack4 MLP	0.717	26.177	14.550	35.326	46.579	-0.495
Stack5 LinearRegression _{global}	0.717	26.905	14.552	35.034	46.852	-0.053
Stack5 Ridge _{global}	0.717	26.910	14.547	35.022	46.818	-0.064
Stack4 LinearRegression _{global}	0.717	26.912	14.590	35.127	46.948	-0.056
Stack4 Ridge _{global}	0.717	26.913	14.584	35.112	46.912	-0.066
Stack5 LinearRegression	0.716	26.247	14.600	35.445	47.345	-0.545
Stack5 Ridge	0.716	26.247	14.599	35.445	47.344	-0.545
Stack1 MLP	0.716	26.257	14.669	35.614	47.371	-0.555
Stack4 LinearRegression	0.715	26.271	14.621	35.498	47.385	-0.553
Stack4 Ridge	0.715	26.271	14.621	35.498	47.385	-0.553
Stack2 GradientBoosting _{global}	0.715	27.001	14.536	34.996	46.544	-0.181
Stack1 Ridge	0.714	26.339	14.689	35.663	47.140	-0.500
Stack1 LinearRegression	0.714	26.339	14.689	35.663	47.140	-0.500
Stack4 GradientBoosting	0.713	26.383	14.607	35.464	47.377	-0.674
Stack2 MLP	0.712	26.401	14.587	35.414	46.924	-0.528
Stack3 LinearRegression _{global}	0.712	27.155	14.647	35.265	47.109	-0.126
Stack3 Ridge _{global}	0.712	27.157	14.640	35.248	47.066	-0.138
Stack2 LinearRegression	0.712	26.429	14.632	35.523	47.316	-0.535
Stack2 Ridge	0.712	26.429	14.632	35.523	47.316	-0.535
Stack3 MLP _{global}	0.711	27.212	14.529	34.979	46.365	-0.705
Stack2 SVR _{global}	0.711	27.212	14.225	34.247	45.732	-2.514
Stack4 SVR _{global}	0.711	27.225	14.249	34.305	46.002	-2.559
Stack5 SVR _{global}	0.710	27.237	14.218	34.232	45.909	-2.549
Stack1 SVR _{global}	0.710	27.240	14.307	34.445	45.502	-2.747
Stack3 MLP	0.709	26.544	14.875	36.114	47.914	-0.401
Stack5 SVR	0.709	26.550	14.241	34.575	47.201	-4.012
Stack5 RandomForest	0.708	26.594	15.177	36.847	48.063	0.095
Stack4 SVR	0.708	26.599	14.320	34.766	47.387	-4.112
Stack3 SVR _{global}	0.706	27.453	14.305	34.441	46.147	-2.626
Stack1 SVR	0.705	26.729	14.400	34.961	46.958	-3.988
Stack2 SVR	0.704	26.784	14.279	34.667	47.082	-4.027
Stack3 LinearRegression	0.703	26.832	14.942	36.276	48.431	-0.664
Stack3 Ridge	0.703	26.832	14.942	36.276	48.431	-0.664
Stack4 RandomForest	0.702	26.897	15.442	37.490	48.861	-0.050
Stack5 GradientBoosting _{global}	0.701	27.690	14.571	35.081	46.490	-0.240
Stack1 GradientBoosting _{global}	0.699	27.740	14.641	35.250	46.608	-0.367
Stack3 GradientBoosting _{global}	0.699	27.770	14.642	35.253	46.471	-0.282
Stack3 SVR	0.696	27.146	14.613	35.477	48.257	-4.409
Stack1 GradientBoosting	0.695	27.176	14.815	35.967	47.547	-0.476
Stack4 GradientBoosting _{global}	0.693	28.026	14.640	35.247	46.441	-0.166
Stack2 RandomForest _{global}	0.690	28.187	15.317	36.876	47.788	0.139
Stack2 GradientBoosting	0.690	27.429	14.710	35.714	47.413	-0.599
Stack2 RandomForest	0.688	27.488	15.562	37.782	48.807	-0.134
Stack3 RandomForest	0.685	27.622	15.914	38.636	49.763	-0.395
Stack1 RandomForest _{global}	0.681	28.578	15.738	37.891	48.764	-0.085
Stack5 RandomForest _{global}	0.679	28.674	15.225	36.656	47.660	0.534
Stack4 RandomForest _{global}	0.673	28.956	15.339	36.929	47.706	0.394
Stack1 RandomForest	0.672	28.193	16.193	39.314	50.293	-0.396
Stack3 RandomForest _{global}	0.667	29.218	15.575	37.499	48.075	0.408

Table A.3

Base Model Comparison for IFN2 (carbono_bruto4)

Model	R^2_{test}	RMSE _{test} (tC)	MAE _{test} (tC)	wMAPE (%)	SMAPE (%)	Bias (tC)
CatBoost_{global}	0.765	18.533	10.402	35.470	69.561	-0.342
LightGBM _{global}	0.762	18.627	10.417	35.522	69.531	-0.313
XGBoost _{global}	0.762	18.644	10.428	35.559	69.353	-0.384
GBDT _{global}	0.759	18.765	10.474	35.716	69.410	-0.407
CatBoost	0.748	19.071	11.036	38.035	72.181	0.034
MLP _{global}	0.747	19.220	10.732	36.597	71.377	-1.309
LightGBM	0.747	19.121	10.858	37.423	71.698	0.042
XGBoost	0.742	19.300	10.962	37.780	71.609	-0.004
BaggedDT _{global}	0.738	19.550	11.216	38.246	71.630	-0.749
RandomForest _{global}	0.734	19.694	10.928	37.264	68.279	-1.503
GBDT	0.732	19.692	11.550	39.809	73.147	0.016
RandomForest	0.729	19.770	11.294	38.925	70.570	0.013
BaggedDT	0.728	19.832	11.349	39.114	70.434	0.182
BayesianNN	0.693	21.069	12.830	44.220	79.896	-0.076
MLP	0.690	21.162	11.419	39.356	76.074	-3.511
BayesianNN _{global}	0.683	21.530	12.827	43.741	78.379	-0.859
SVR _{global}	0.577	24.861	12.305	41.961	75.324	-7.293
SVR	0.566	25.034	12.818	44.177	77.708	-6.476

A.2. IFN2 as explanatory variable for carbono_bruto4 (tC)**A.2.1. Base models**

Table A.3 shows the performance summary of the models for predicting the carbon variable in tonnes (carbono_bruto4) using the dataset that employs IFN2 as the explanatory inventory.

A.2.2. Stacking models

Table A.4 shows the performance summary of the stacking models for predicting the carbon variable in tonnes (carbono_bruto4) using the dataset that employs IFN2 as the explanatory inventory.

Table A.4

Stacking Model Comparison for IFN2 (carbono_bruto4)

Model	R^2_{test}	RMSE _{test} (tC)	MAE _{test} (tC)	wMAPE (%)	SMAPE (%)	Bias (tC)
Stack5 MLP_{global}	0.768	18.405	10.210	34.817	68.129	-0.195
Stack5 LinearRegression _{global}	0.768	18.418	10.306	35.144	69.309	-0.169
Stack5 Ridge _{global}	0.768	18.418	10.306	35.143	69.320	-0.170
Stack4 MLP _{global}	0.767	18.433	10.229	34.881	68.095	0.030
Stack4 LinearRegression _{global}	0.767	18.435	10.297	35.112	69.291	-0.232
Stack4 Ridge _{global}	0.767	18.435	10.296	35.111	69.288	-0.233
Stack3 MLP _{global}	0.767	18.450	10.184	34.726	67.782	-0.295
Stack5 GradientBoosting _{global}	0.767	18.457	10.224	34.863	68.228	-0.393
Stack3 Ridge _{global}	0.767	18.466	10.318	35.185	69.212	-0.277
Stack3 LinearRegression _{global}	0.767	18.466	10.318	35.185	69.212	-0.277
Stack4 GradientBoosting _{global}	0.766	18.485	10.216	34.837	68.125	-0.407
Stack2 LinearRegression _{global}	0.765	18.521	10.333	35.236	69.359	-0.252
Stack2 Ridge _{global}	0.765	18.521	10.333	35.236	69.360	-0.253
Stack3 GradientBoosting _{global}	0.765	18.529	10.230	34.886	68.141	-0.463
Stack2 MLP _{global}	0.765	18.531	10.285	35.071	68.797	-0.207
Stack5 SVR _{global}	0.763	18.595	10.155	34.628	68.878	-1.814
Stack4 SVR _{global}	0.763	18.601	10.169	34.676	68.822	-1.826
Stack1 MLP _{global}	0.763	18.602	10.261	34.989	67.896	-0.730
Stack1 Ridge _{global}	0.763	18.606	10.377	35.385	69.330	-0.345
Stack1 LinearRegression _{global}	0.763	18.606	10.377	35.385	69.330	-0.345
Stack2 GradientBoosting _{global}	0.763	18.616	10.297	35.114	68.617	-0.412
Stack3 SVR _{global}	0.762	18.643	10.193	34.758	68.689	-1.883
Stack1 GradientBoosting _{global}	0.761	18.676	10.316	35.177	68.372	-0.484
Stack2 SVR _{global}	0.761	18.678	10.222	34.856	68.947	-1.763
Stack1 SVR _{global}	0.758	18.821	10.260	34.986	68.943	-2.023
Stack5 RandomForest _{global}	0.756	18.879	10.606	36.167	66.996	0.138
Stack4 RandomForest _{global}	0.756	18.886	10.704	36.501	67.217	0.084
Stack4 MLP	0.755	18.828	10.636	36.658	70.675	0.274
Stack4 Ridge	0.754	18.860	10.734	36.997	71.361	0.044
Stack4 LinearRegression	0.754	18.860	10.734	36.997	71.364	0.044
Stack5 GradientBoosting	0.753	18.880	10.692	36.852	70.603	-0.022
Stack5 MLP	0.753	18.890	10.632	36.643	71.389	0.045
Stack5 Ridge	0.753	18.893	10.746	37.038	71.326	0.016
Stack5 LinearRegression	0.753	18.893	10.747	37.039	71.327	-0.016
Stack3 MLP	0.753	18.896	10.746	37.036	71.008	0.081
Stack4 GradientBoosting	0.752	18.921	10.654	36.720	70.494	0.029
Stack2 MLP	0.751	18.954	10.765	37.102	70.755	-0.001
Stack3 Ridge	0.751	18.977	10.873	37.473	71.697	0.076
Stack3 LinearRegression	0.751	18.978	10.873	37.474	71.695	0.076
Stack2 GradientBoosting	0.751	18.978	10.769	37.116	70.909	0.002
Stack4 SVR	0.750	18.988	10.523	36.267	72.240	-2.179
Stack2 Ridge	0.750	18.989	10.793	37.200	71.359	-0.048
Stack2 LinearRegression	0.750	18.989	10.793	37.200	71.357	-0.048
Stack5 SVR	0.750	18.993	10.530	36.293	72.199	-2.129
Stack3 GradientBoosting	0.749	19.027	10.783	37.164	70.757	0.076
Stack1 MLP	0.749	19.043	10.736	37.001	70.468	-0.031
Stack1 Ridge	0.749	19.048	10.777	37.145	71.433	-0.008
Stack1 LinearRegression	0.749	19.048	10.777	37.145	71.434	-0.008
Stack3 SVR	0.747	19.116	10.656	36.726	72.643	-2.243
Stack2 SVR	0.747	19.116	10.602	36.542	72.634	-2.230
Stack1 GradientBoosting	0.746	19.174	10.778	37.146	70.768	-0.046
Stack3 RandomForest _{global}	0.745	19.286	10.840	36.965	67.152	-0.105
Stack1 SVR	0.745	19.187	10.596	36.521	72.624	-2.197
Stack2 RandomForest _{global}	0.741	19.455	10.979	37.440	69.244	-0.110
Stack5 RandomForest	0.734	19.588	11.135	38.377	68.643	0.589
Stack1 RandomForest _{global}	0.734	19.711	11.235	38.313	68.420	-0.278
Stack3 RandomForest	0.727	19.848	11.483	39.578	69.750	0.543
Stack4 RandomForest	0.727	19.848	11.194	38.580	68.556	0.550
Stack2 RandomForest	0.720	20.098	11.538	39.768	71.966	0.196
Stack1 RandomForest	0.713	20.375	11.767	40.555	70.731	0.068

Table A.5

Base Model Comparison for IFN3 (c4)

Model	R^2_{test}	RMSE _{test} (tC/ha)	MAE _{test} (tC/ha)	wMAPE (%)	SMAPE (%)	Bias (tC/ha)
LightGBM	0.858	17.816	9.246	24.609	37.327	0.017
XGBoost	0.851	18.232	9.203	24.494	37.033	-0.021
CatBoost	0.849	18.385	9.746	25.941	40.139	-0.114
GBDT	0.843	18.718	9.893	26.330	40.046	0.005
MLP	0.840	18.947	9.296	24.743	36.827	-3.555
GBDT _{global}	0.839	19.408	9.559	25.792	38.647	0.378
XGBoost _{global}	0.839	19.424	9.548	25.763	39.032	0.206
CatBoost _{global}	0.838	19.500	9.433	25.451	39.540	0.056
LightGBM _{global}	0.836	19.580	9.602	25.909	39.178	0.237
MLP _{global}	0.833	19.791	10.008	27.005	44.150	0.107
BaggedDT	0.820	20.085	10.699	28.476	40.585	0.098
BaggedDT _{global}	0.800	21.649	11.058	29.838	44.515	1.538
RandomForest	0.796	21.353	10.839	28.849	40.827	-0.092
RandomForest _{global}	0.789	22.223	11.109	29.974	41.901	1.399
BayesianNN	0.774	22.507	12.332	32.823	55.074	0.201
BayesianNN _{global}	0.743	24.561	12.643	34.113	56.226	0.572
SVR	0.737	24.279	9.877	26.289	37.234	-3.632
SVR _{global}	0.613	30.113	11.708	31.590	42.962	-4.778

A.3. IFN3 as explanatory variable for c4 (tC/ha)**A.3.1. Base models**

Once the models are trained, global parameters such as R^2 , RMSE and MAE are presented in Table A.5.

A.3.2. Stacking models

Table A.6 shows the performance summary of the stacking models for predicting the carbon variable in tC/ha using the dataset that employs IFN3 as the explanatory inventory.

Table A.6
Stacking Model Comparison for IFN3 (c4)

Model	R^2_{test}	RMSE _{test} (tC/ha)	MAE _{test} (tC/ha)	wMAPE (%)	SMAPE (%)	Bias (tC/ha)
Stack4 GradientBoosting	0.866	17.303	9.134	24.310	37.215	-0.009
Stack5 GradientBoosting	0.865	17.369	9.037	24.052	36.846	-0.007
Stack2 GradientBoosting	0.863	17.537	9.097	24.211	37.106	0.005
Stack5 MLP	0.862	17.572	8.950	23.822	36.332	-0.335
Stack5 LinearRegression	0.861	17.644	9.049	24.083	37.004	-0.013
Stack5 Ridge	0.861	17.646	9.050	24.087	37.003	-0.012
Stack4 MLP	0.861	17.647	9.147	24.346	36.813	0.171
Stack2 MLP	0.860	17.704	9.038	24.054	36.404	-0.294
Stack4 LinearRegression	0.860	17.718	9.190	24.459	37.349	0.011
Stack4 Ridge	0.860	17.720	9.191	24.463	37.351	0.011
Stack1 GradientBoosting	0.859	17.740	9.273	24.681	37.369	0.029
Stack2 LinearRegression	0.859	17.767	9.097	24.213	36.750	-0.004
Stack2 Ridge	0.859	17.773	9.107	24.240	36.785	-0.004
Stack5 SVR	0.858	17.816	8.879	23.631	37.125	-1.871
Stack5 RandomForest	0.858	17.819	9.445	25.139	37.517	0.362
Stack1 MLP	0.858	17.834	9.264	24.657	37.281	0.119
Stack4 SVR	0.858	17.851	9.011	23.984	37.554	-1.902
Stack1 LinearRegression	0.857	17.873	9.244	24.604	36.837	0.023
Stack1 Ridge	0.857	17.875	9.245	24.606	36.831	0.023
Stack2 SVR	0.856	17.954	8.935	23.781	37.175	-1.851
Stack3 GradientBoosting	0.855	18.026	9.534	25.375	38.431	0.002
Stack1 SVR	0.854	18.053	9.114	24.257	37.328	-1.845
Stack3 MLP	0.852	18.171	9.500	25.285	37.589	0.001
Stack4 RandomForest	0.852	18.199	9.748	25.946	38.468	0.330
Stack3 LinearRegression	0.850	18.309	9.580	25.498	38.816	0.007
Stack3 Ridge	0.850	18.316	9.581	25.500	38.784	0.008
Stack3 RandomForest	0.849	18.375	10.159	27.039	39.787	0.336
Stack5 MLP _{global}	0.848	18.892	9.178	24.763	36.606	0.347
Stack2 MLP _{global}	0.847	18.943	9.332	25.180	37.530	0.382
Stack3 SVR	0.846	18.552	9.397	25.010	38.601	-2.106
Stack4 MLP _{global}	0.845	19.067	9.180	24.770	36.741	0.146
Stack5 Ridge _{global}	0.844	19.100	9.252	24.965	38.537	0.140
Stack5 LinearRegression _{global}	0.844	19.102	9.253	24.966	38.597	0.131
Stack2 Ridge _{global}	0.844	19.115	9.355	25.241	38.497	0.193
Stack2 LinearRegression _{global}	0.844	19.117	9.356	25.244	38.525	0.191
Stack4 Ridge _{global}	0.843	19.154	9.278	25.033	38.451	0.184
Stack4 LinearRegression _{global}	0.843	19.156	9.277	25.030	38.505	0.173
Stack3 Ridge _{global}	0.843	19.187	9.305	25.107	38.570	0.218
Stack3 LinearRegression _{global}	0.843	19.188	9.304	25.104	38.641	0.205
Stack3 MLP _{global}	0.842	19.216	9.188	24.790	36.845	-0.169
Stack5 SVR _{global}	0.840	19.367	9.101	24.557	40.002	-2.063
Stack2 SVR _{global}	0.840	19.369	9.185	24.783	39.985	-2.057
Stack4 SVR _{global}	0.839	19.400	9.116	24.597	39.978	-2.045
Stack3 SVR _{global}	0.839	19.423	9.137	24.655	40.013	-2.018
Stack2 RandomForest	0.839	18.988	9.887	26.314	38.705	0.293
Stack1 MLP _{global}	0.839	19.436	9.564	25.804	37.443	0.770
Stack3 GradientBoosting _{global}	0.837	19.547	9.290	25.067	37.493	0.310
Stack1 LinearRegression _{global}	0.837	19.564	9.582	25.854	38.721	0.337
Stack1 Ridge _{global}	0.837	19.565	9.582	25.854	38.714	0.339
Stack1 RandomForest	0.835	19.224	10.450	27.814	40.024	0.233
Stack5 GradientBoosting _{global}	0.833	19.768	9.291	25.069	37.539	0.270
Stack1 GradientBoosting _{global}	0.832	19.820	9.600	25.903	38.177	0.508
Stack1 SVR _{global}	0.832	19.852	9.388	25.330	39.687	-1.927
Stack4 GradientBoosting _{global}	0.830	19.940	9.344	25.213	37.491	0.304
Stack2 GradientBoosting _{global}	0.826	20.206	9.499	25.631	38.253	0.331
Stack5 RandomForest _{global}	0.820	20.540	9.871	26.635	38.299	0.692
Stack3 RandomForest _{global}	0.807	21.266	10.271	27.713	39.177	0.574
Stack4 RandomForest _{global}	0.804	21.415	10.123	27.314	38.710	0.763
Stack2 RandomForest _{global}	0.802	21.562	10.342	27.905	39.864	0.540
Stack1 RandomForest _{global}	0.792	22.090	10.962	29.576	40.925	0.731

Table A.7

Base Model Comparison for IFN3 (carbono_bruto4)

Model	R^2_{test}	RMSE _{test} (tC)	MAE _{test} (tC)	wMAPE (%)	SMAPE (%)	Bias (tC)
LightGBM	0.898	10.641	4.687	27.801	110.183	-0.007
CatBoost	0.896	10.737	4.789	28.405	110.952	-0.059
XGBoost	0.895	10.761	4.713	27.957	109.669	-0.008
CatBoost _{global}	0.889	10.931	4.812	28.723	111.600	-0.059
GBDT	0.888	11.134	5.044	29.920	111.422	-0.042
MLP	0.885	11.287	4.803	28.488	111.603	-0.544
LightGBM _{global}	0.884	11.153	4.863	29.023	111.679	-0.073
XGBoost _{global}	0.883	11.230	4.859	28.999	111.190	-0.036
GBDT _{global}	0.881	11.327	4.937	29.463	111.656	-0.002
MLP _{global}	0.879	11.431	5.122	30.573	114.182	-0.326
BaggedDT	0.873	11.830	5.183	30.745	99.986	-0.135
RandomForest	0.870	11.965	5.217	30.944	109.139	-0.239
BaggedDT _{global}	0.866	11.997	5.410	32.287	110.035	0.589
RandomForest _{global}	0.865	12.053	5.330	31.814	105.593	0.434
BayesianNN	0.836	13.455	6.773	40.173	120.931	0.010
BayesianNN _{global}	0.824	13.774	7.040	42.020	121.650	0.265
SVR	0.810	14.504	5.269	31.256	112.607	-2.176
SVR _{global}	0.730	17.033	6.154	36.728	114.785	-2.987

A.4. IFN3 as explanatory variable for carbono_bruto4 (tC)**A.4.1. Base models**

Table A.7 shows the performance summary of the models for predicting the carbon variable in tonnes (carbono_bruto4) using the dataset that employs IFN3 as the explanatory inventory.

A.4.2. Stacking models

Table A.8 shows the performance summary of the stacking models for predicting the carbon variable in tonnes (carbono_bruto4) using the dataset that employs IFN3 as the explanatory inventory.

Table A.8

Stacking Model Comparison for IFN3 (carbono_bruto4)

Model	R^2_{test}	RMSE _{test} (tC)	MAE _{test} (tC)	wMAPE (%)	SMAPE (%)	Bias (tC)
Stack4 MLP	0.902	10.407	4.517	26.795	107.868	0.049
Stack5 GradientBoosting	0.901	10.433	4.540	26.930	109.331	0.030
Stack5 MLP	0.901	10.447	4.454	26.417	107.446	-0.368
Stack4 GradientBoosting	0.901	10.450	4.541	26.936	109.286	0.037
Stack5 LinearRegression	0.900	10.492	4.624	27.426	110.839	0.025
Stack5 Ridge	0.900	10.494	4.624	27.428	110.757	0.022
Stack4 LinearRegression	0.900	10.511	4.648	27.570	110.932	0.020
Stack4 Ridge	0.900	10.511	4.648	27.570	110.941	0.020
Stack2 MLP	0.899	10.540	4.581	27.173	108.573	-0.035
Stack2 LinearRegression	0.899	10.572	4.651	27.590	110.865	0.011
Stack3 MLP	0.899	10.572	4.611	27.348	106.609	0.170
Stack2 Ridge	0.899	10.573	4.654	27.604	110.866	0.010
Stack2 GradientBoosting	0.899	10.582	4.610	27.343	109.772	0.004
Stack1 MLP	0.898	10.617	4.579	27.162	107.666	-0.163
Stack1 LinearRegression	0.898	10.625	4.667	27.682	111.022	0.011
Stack1 Ridge	0.898	10.625	4.667	27.682	111.022	0.011
Stack5 SVR	0.898	10.638	4.541	26.938	110.810	-0.758
Stack4 SVR	0.897	10.655	4.556	27.027	111.084	-0.788
Stack1 GradientBoosting	0.897	10.660	4.620	27.406	109.313	0.021
Stack3 GradientBoosting	0.897	10.671	4.637	27.506	109.803	0.028
Stack3 LinearRegression	0.897	10.674	4.757	28.218	111.578	0.014
Stack3 Ridge	0.897	10.675	4.757	28.217	111.563	0.013
Stack2 SVR	0.896	10.706	4.580	27.167	110.761	-0.744
Stack5 RandomForest	0.895	10.755	4.721	28.006	96.178	0.266
Stack4 RandomForest	0.895	10.782	4.775	28.322	98.323	0.216
Stack1 SVR	0.895	10.787	4.613	27.364	110.982	-0.740
Stack3 SVR	0.893	10.865	4.672	27.712	111.457	-0.870
Stack4 MLP _{global}	0.890	10.862	4.621	27.580	109.222	0.226
Stack5 MLP _{global}	0.890	10.866	4.586	27.369	107.939	0.041
Stack3 MLP _{global}	0.890	10.880	4.544	27.119	106.128	-0.009
Stack4 LinearRegression _{global}	0.890	10.890	4.741	28.298	111.854	-0.075
Stack4 Ridge _{global}	0.890	10.891	4.741	28.297	111.872	-0.075
Stack5 GradientBoosting _{global}	0.890	10.905	4.612	27.525	109.827	-0.006
Stack5 LinearRegression _{global}	0.889	10.908	4.747	28.330	112.364	-0.113
Stack5 Ridge _{global}	0.889	10.909	4.747	28.330	112.377	-0.113
Stack3 LinearRegression _{global}	0.889	10.921	4.766	28.443	112.073	-0.047
Stack3 Ridge _{global}	0.889	10.921	4.766	28.443	112.073	-0.047
Stack3 GradientBoosting _{global}	0.888	10.959	4.626	27.612	109.664	0.036
Stack4 GradientBoosting _{global}	0.888	10.978	4.625	27.605	109.591	0.007
Stack3 RandomForest	0.887	11.157	4.967	29.463	98.766	0.152
Stack2 RandomForest	0.887	11.165	4.916	29.157	98.358	0.133
Stack4 SVR _{global}	0.887	11.053	4.682	27.941	112.576	-0.961
Stack5 SVR _{global}	0.886	11.056	4.687	27.976	112.536	-0.956
Stack2 MLP _{global}	0.886	11.058	4.709	28.105	109.662	0.075
Stack2 LinearRegression _{global}	0.886	11.065	4.803	28.663	112.597	-0.094
Stack2 Ridge _{global}	0.886	11.065	4.802	28.662	112.594	-0.094
Stack3 SVR _{global}	0.886	11.088	4.701	28.059	112.546	-0.948
Stack1 MLP _{global}	0.885	11.102	4.680	27.934	107.602	-0.133
Stack2 GradientBoosting _{global}	0.885	11.108	4.729	28.224	110.752	-0.005
Stack1 Ridge _{global}	0.885	11.111	4.833	28.842	112.219	-0.040
Stack1 LinearRegression _{global}	0.885	11.111	4.833	28.842	112.219	-0.040
Stack1 GradientBoosting _{global}	0.884	11.171	4.732	28.242	109.678	0.048
Stack2 SVR _{global}	0.883	11.220	4.745	28.321	112.576	-0.971
Stack1 SVR _{global}	0.882	11.271	4.760	28.409	112.572	-0.951
Stack1 RandomForest	0.881	11.483	5.127	30.409	95.521	0.089
Stack5 RandomForest _{global}	0.880	11.347	4.848	28.932	96.971	0.259
Stack4 RandomForest _{global}	0.879	11.429	4.919	29.356	96.704	0.192
Stack3 RandomForest _{global}	0.875	11.586	5.023	29.976	95.997	0.204
Stack2 RandomForest _{global}	0.873	11.692	5.131	30.625	102.247	0.209
Stack1 RandomForest _{global}	0.858	12.368	5.383	32.129	94.456	0.154

Table A.9
Global Base Models (c4)

Model	R^2_{test}	RMSE _{test} (tC/ha)	MAE _{test} (tC/ha)	wMAPE (%)	SMAPE (%)	Bias (tC/ha)
LightGBM_{global}	0.787	22.767	11.650	29.978	42.370	0.134
XGBoost _{global}	0.784	22.952	11.590	29.824	42.204	0.055
CatBoost _{global}	0.783	22.990	11.607	29.866	42.904	-0.021
GBDT _{global}	0.783	23.014	11.658	29.998	41.987	0.169
MLP _{global}	0.771	23.606	12.287	31.617	46.726	0.177
BaggedDT _{global}	0.740	25.142	13.021	33.506	47.116	0.678
RandomForest _{global}	0.732	25.547	12.908	33.214	44.513	0.232
BayesianNN _{global}	0.678	28.021	14.689	37.797	56.389	0.047
SVR _{global}	0.551	33.065	13.708	35.272	45.693	-6.204

Table A.10
Global Stacking Models (c4)

Model	R^2_{test}	RMSE _{test} (tC/ha)	MAE _{test} (tC/ha)	wMAPE (%)	SMAPE (%)	Bias (tC/ha)
Stack5 MLP_{global}	0.794	22.387	11.321	29.132	40.529	0.238
Stack2 MLP _{global}	0.794	22.405	11.403	29.341	41.069	0.176
Stack4 MLP _{global}	0.792	22.527	11.314	29.112	40.574	-0.045
Stack2 LinearRegression _{global}	0.791	22.565	11.429	29.409	41.712	0.069
Stack2 Ridge _{global}	0.791	22.566	11.429	29.409	41.696	0.069
Stack5 LinearRegression _{global}	0.791	22.571	11.387	29.299	41.921	0.057
Stack5 Ridge _{global}	0.791	22.572	11.384	29.293	41.871	0.058
Stack4 Ridge _{global}	0.790	22.601	11.414	29.371	41.858	0.083
Stack4 LinearRegression _{global}	0.790	22.602	11.416	29.375	41.905	0.081
Stack1 MLP _{global}	0.790	22.635	11.584	29.808	41.096	0.507
Stack3 LinearRegression _{global}	0.788	22.735	11.456	29.477	42.051	0.072
Stack3 Ridge _{global}	0.788	22.735	11.453	29.471	41.991	0.074
Stack1 LinearRegression _{global}	0.787	22.769	11.606	29.864	41.939	0.145
Stack1 Ridge _{global}	0.787	22.770	11.606	29.864	41.933	0.145
Stack3 MLP _{global}	0.787	22.776	11.338	29.175	40.679	-0.384
Stack2 SVR _{global}	0.786	22.853	11.214	28.856	42.299	-2.241
Stack5 SVR _{global}	0.785	22.864	11.162	28.721	42.381	-2.258
Stack4 SVR _{global}	0.785	22.875	11.183	28.775	42.403	-2.252
Stack3 SVR _{global}	0.783	22.996	11.218	28.866	42.483	-2.263
Stack1 SVR _{global}	0.781	23.113	11.369	29.253	42.029	-2.257
Stack2 GradientBoosting _{global}	0.779	23.183	11.527	29.661	41.592	0.125
Stack3 GradientBoosting _{global}	0.779	23.211	11.445	29.450	41.108	0.072
Stack5 GradientBoosting _{global}	0.777	23.284	11.417	29.378	41.143	0.064
Stack1 GradientBoosting _{global}	0.776	23.334	11.630	29.926	41.572	0.156
Stack4 GradientBoosting _{global}	0.773	23.532	11.477	29.531	41.095	0.115
Stack5 RandomForest _{global}	0.761	24.147	12.027	30.947	42.068	0.628
Stack2 RandomForest _{global}	0.755	24.446	12.345	31.766	43.055	0.378
Stack4 RandomForest _{global}	0.749	24.730	12.223	31.452	42.332	0.614
Stack3 RandomForest _{global}	0.748	24.777	12.407	31.924	42.760	0.507
Stack1 RandomForest _{global}	0.745	24.907	12.885	33.155	44.082	0.402

A.5. IFN2+IFN3 as explanatory variables for c4 (tC/ha)

A.5.1. Base models

Once the models are trained, global parameters such as R^2 , RMSE and MAE are presented in Table A.9.

A.5.2. Stacking models

Table A.10 shows the performance summary of the stacking models for predicting the carbon variable in tC/ha using the dataset that employs IFN2 and IFN3 as explanatory inventories.

Table A.11

Global Base Models (carbono_bruto4)

Model	R^2_{test}	RMSE _{test} (tC)	MAE _{test} (tC)	wMAPE (%)	SMAPE (%)	Bias (tC)
CatBoost_{global}	0.845	13.846	6.615	31.789	98.044	-0.150
LightGBM _{global}	0.841	14.006	6.654	31.977	98.088	-0.150
XGBoost _{global}	0.840	14.054	6.655	31.980	97.699	-0.148
GBDT _{global}	0.838	14.159	6.722	32.304	98.033	-0.132
MLP _{global}	0.832	14.410	6.931	33.311	100.379	-0.643
BaggedDT _{global}	0.821	14.858	7.282	34.995	97.651	0.158
RandomForest _{global}	0.819	14.950	7.135	34.291	93.561	-0.191
BayesianNN _{global}	0.775	16.674	8.906	42.802	107.697	-0.097
SVR _{global}	0.679	19.897	8.137	39.106	102.061	-4.376

Table A.12

Global Stacking Models (carbono_bruto4)

Model	R^2_{test}	RMSE _{test} (tC)	MAE _{test} (tC)	wMAPE (%)	SMAPE (%)	Bias (tC)
Stack5 MLP_{global}	0.847	13.756	6.399	30.754	95.102	-0.035
Stack4 MLP _{global}	0.847	13.766	6.429	30.898	95.960	0.163
Stack4 LinearRegression _{global}	0.846	13.782	6.533	31.394	98.129	-0.126
Stack4 Ridge _{global}	0.846	13.782	6.533	31.394	98.140	-0.126
Stack3 MLP _{global}	0.846	13.783	6.362	30.576	93.763	-0.101
Stack5 LinearRegression _{global}	0.846	13.785	6.539	31.426	98.481	-0.131
Stack5 Ridge _{global}	0.846	13.785	6.539	31.426	98.493	-0.131
Stack5 GradientBoosting _{global}	0.846	13.799	6.421	30.860	96.413	-0.131
Stack3 LinearRegression _{global}	0.846	13.812	6.556	31.506	98.252	-0.121
Stack3 Ridge _{global}	0.846	13.812	6.556	31.506	98.252	-0.121
Stack4 GradientBoosting _{global}	0.845	13.850	6.428	30.891	96.220	-0.126
Stack3 GradientBoosting _{global}	0.844	13.860	6.433	30.917	96.274	-0.125
Stack2 LinearRegression _{global}	0.843	13.913	6.586	31.650	98.655	-0.145
Stack2 Ridge _{global}	0.843	13.913	6.586	31.650	98.653	-0.145
Stack2 MLP _{global}	0.843	13.913	6.507	31.270	96.485	-0.016
Stack5 SVR _{global}	0.843	13.940	6.450	30.999	98.458	-1.233
Stack4 SVR _{global}	0.843	13.941	6.451	31.002	98.467	-1.240
Stack1 MLP _{global}	0.842	13.968	6.480	31.140	94.798	-0.326
Stack1 Ridge _{global}	0.842	13.974	6.620	31.816	98.389	-0.138
Stack1 LinearRegression _{global}	0.842	13.974	6.620	31.816	98.389	-0.138
Stack2 GradientBoosting _{global}	0.842	13.977	6.524	31.355	97.165	-0.136
Stack3 SVR _{global}	0.842	13.977	6.472	31.103	98.404	-1.249
Stack1 GradientBoosting _{global}	0.840	14.037	6.532	31.393	96.359	-0.123
Stack2 SVR _{global}	0.840	14.064	6.511	31.290	98.508	-1.226
Stack1 SVR _{global}	0.838	14.153	6.533	31.398	98.503	-1.297
Stack5 RandomForest _{global}	0.836	14.218	6.705	32.220	87.305	0.220
Stack4 RandomForest _{global}	0.835	14.266	6.784	32.603	87.196	0.157
Stack3 RandomForest _{global}	0.829	14.522	6.898	33.152	86.696	0.104
Stack2 RandomForest _{global}	0.826	14.652	7.017	33.722	91.605	0.106
Stack1 RandomForest _{global}	0.815	15.130	7.270	34.939	86.061	0.015

A.6. IFN2+IFN3 as explanatory variables for carbono_bruto4 (tC)**A.6.1. Base models**

Table A.11 shows the performance summary of the models for predicting the carbon variable in tonnes using the dataset that employs IFN2 and IFN3 as explanatory inventories.

A.6.2. Stacking models

Table A.12 shows the performance summary of the stacking models for predicting the carbon variable in tonnes using the dataset that employs IFN2 and IFN3 as explanatory inventories.